MPGU-Net: Multimodel Physics-Guided Network for Nonlinear Hyperspectral Unmixing

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***Abstract*—Hyperspectral unmixing remains a challenging problem in complex scenes, where the linear model cannot describe the spectral reflectance of mixtures of pure materials. Although nonlinear mixing models have been proposed to address this challenge, existing methods are either supervised, i.e., require endmembers of the dataset as prior information, or are suitable only for datasets containing either pure pixels or near-pure pixels. When pure pixels are unavailable in the dataset, a dataset simplex has to be reconstructed in order to generate virtual endmembers. Unfortunately, the popular minimum simplex volume constraint (MSVC) often used for linear datasets cannot be directly applied to nonlinear datasets. To tackle this challenge, in this work, we propose a novel strategy called linearizing space transformation (LST) that estimates the linear component of a nonlinear dataset by analytically inverting the nonlinear dataset. MSVC can then be applied to this linear component to estimate the endmembers of the nonlinear dataset. LST can be applied to nonlinear datasets that can be expressed as an explicit combination of linear and nonlinear components. The polynomial mixing models are a class of models that fulfill this criterion. In this work, we theoretically demonstrate the effectiveness of the proposed strategy for two popular mixing models from the literature, i.e., the polynomial postnonlinear mixing model (PPNM) and the multilinear mixing model (MLM). In order to estimate endmembers, abundance maps, and nonlinear parameters of the mixing models, an end-to-end deep network called MPGU-Net is proposed. The network consists of three modules: 1) a variational autoencoder (VAE)-based endmember generation network for robust spectral signature estimation; 2) an abundance estimation network enhanced by a convolutional block attention module (CBAM) for accurately estimating abundance maps; and 3) a nonlinearity parameter estimation network for estimating pixelwise nonlinear interaction processes. In contrast to existing methods that model nonlinearity through a nonlinear activation function in the last layer of the network, our approach models nonlinearity explicitly through the nonlinear mixing equation. LST is not a direct component of the MPGU-Net framework, but is incorporated into the loss function, enabling more accurate estimation of endmembers and fractional abundances. By jointly optimizing reconstruction consistency, geometric constraints, endmember consistency, and variational regularization, our MPGU-Net achieves high-accuracy unmixing with strong physical interpretability. Extensive experiments on both synthetic and real hyperspectral datasets demonstrate that our method significantly outperforms existing state-of-the-art (linear and nonlinear) unmixing approaches in terms of reconstruction accuracy, endmember estimation fidelity, and abundance visualization. We release our code at https://github.com/xuanwentao**

***Index Terms*—Linearizing space transformation (LST), multilinear mixing model (MLM), multimodel physics-guided unmixing network, postnonlinear mixing model (PPNM).**

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I. Introduction

REMOTE sensing is a powerful tool in Earth observation that is widely used for monitoring changes in land use [1], mineral exploration [2], natural disasters [3], and climate change [4]. Among its various applications, the efficient computational processing of multispectral and hyperspectral images, formed by capturing data across multiple spectral bands, is of particular importance. Hyperspectral imaging in particular has attracted significant attention due to its ability to capture detailed spectral information at each pixel, making it highly effective for tasks such as super-resolution [5], target detection [6], classification [7], and anomaly detection [8]. Despite these advantages, hyperspectral data present inherent challenges [9]. Due to the limited spatial resolution of hyperspectral sensors and the complexity of surface materials, each pixel often contains a mixture of multiple pure spectral signatures, known as endmembers. This phenomenon complicates the analysis of hyperspectral images. To address this, hyperspectral unmixing [10], [11], [12], [13] has emerged as a critical technique, aiming to decompose each mixed pixel into a set of endmembers and their corresponding fractional abundances. This process reveals the underlying physical composition of the scene and enhances the interpretability of hyperspectral data.

A. Linear and Nonlinear Unmixing Methods

Hyperspectral unmixing techniques have been primarily developed under two main scenarios: the linear mixing assumption [14], [15], [16] and the nonlinear mixing assumption [10], [17], [18]. The linear mixing model (LMM) assumes that each pixel spectrum can be represented as a linear combination of pure material spectra, referred to as endmembers, weighted by their corresponding fractional abundances.

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This model offers mathematical simplicity, interpretability, and computational efficiency. This model has been widely adopted by numerous hyperspectral unmixing approaches, particularly for scenes in which materials are spatially well-separated. In these scenarios, each incident light ray interacts with a single pure material within the instantaneous field of view before reaching the sensor.

When the pure pixel assumption holds, geometrical methods are typically employed. These methods leverage the geometric property that, under the LMM assumption, all mixed pixels are confined within a simplex whose vertices correspond to the endmembers. Consequently, the unmixing task reduces to identifying these simplex vertices. Well-established algorithms such as vertex component analysis (VCA) [19], N-FINDR [20], [21], and successive projection algorithm (SPA) [22] adopt a maximum volume criterion, aiming to find the simplex of maximum volume that fits the data, which is a reasonable strategy when pure pixels are present.

In contrast, in many real-world scenarios—especially those involving low spatial resolution or highly mixed environments—the pure pixel assumption may not hold. In such cases, geometrical methods may yield inaccurate results. To address this, minimum-volume-based methods have been proposed. These methods aim to identify the smallest simplex that encloses all data points, with virtual endmembers located at its corners. Representative methods include minimum volume constrained nonnegative matrix factorization (MVC-NMF) [23], minimum volume simplex analysis (MVSA) [24], and simplex identification via variable splitting and augmented Lagrangian (SISAL) [25]. Beyond these geometrical approaches, alternative linear unmixing techniques employ statistical inference or optimization frameworks. Bayesian unmixing methods [26], [27], [28], for instance, incorporate prior knowledge about abundance distributions to enhance estimation accuracy. Sparse regression methods [29], [30], [31], [32] exploit the observation that individual pixels typically contain only a limited subset of potential endmembers, thereby enforcing sparsity in the resulting abundance maps. Another prominent approach is nonnegative matrix factorization (NMF) [33], which decomposes the data matrix into a product of two low-rank nonnegative matrices. NMF-based methods [34], [35] are often supplemented with additional constraints to enhance the physical interpretability of the results. In addition, heuristic optimization strategies, including swarm intelligence and evolutionary algorithms, have also been explored for LMM-based unmixing. These methods perform population-based stochastic searches for endmembers and abundances and provide flexible alternatives when the objective function is highly nonconvex or difficult to optimize analytically. Typical algorithms include high-dimensional double-swarm particle swarm optimization (HDPSO) [36], multilevel reweighted sparse unmixing method using superpixel segmentation and particle swarm optimization (MRSUPSO) [37], sparse unmixing via comprehensive learning-based particle swarm optimization (SUCPSO) [38].

However, in many real-world scenarios such as vegetation-covered regions, urban areas, and mineral-rich surfaces, the assumptions of the LMM are often violated because the incident light interacts with multiple pure materials before reaching the sensor. Furthermore, atmospheric effects cause complex spectral distortions that the LMM cannot adequately capture. This results in inaccurate abundance estimates and poor spectral reconstructions when linear unmixing methods are applied under such conditions.

To address these limitations, various nonlinear unmixing methods have been proposed. Among the earliest and most extensively studied are bilinear models [39], [40]. These models assume that incident light interacts with at most two pure materials before reaching the sensor. These models augment the linear mixture with interaction terms representing pairwise multiplicative effects between endmembers. Representative examples include the generalized bilinear model (GBM) and the Fan model [41], [42]. Another prominent family is given by polynomial models such as the polynomial postnonlinear mixing model (PPNM), [43], [44], [45], which applies a nonlinear transformation (typically a second- or third-order polynomial) to the linear mixture. This flexible structure enables the modeling of smooth, weak nonlinearities commonly arising from surface scattering or sensor-induced distortions. Multilinear models, including the multilinear mixing model (MLM), further generalize bilinear models by considering recursive or higher order interactions of the incident light with pure materials. These models [46], [47], [48], [49] are particularly relevant in complex scenes where photons undergo multiple material interactions before reaching the sensor, such as in densely vegetated areas. Radiative transfer models are among the most advanced nonlinear mixing models [10]. They are often used to reconstruct the reflectance spectra of intimately mixed mineral powders. In remote sensing applications, a simplified version of the Hapke model [50], [51] is commonly employed to predict the composition of intimate powder mixtures.

In addition to parametric approaches, kernel-based methods offer a nonparametric framework for modeling nonlinearities. These techniques [52], [53], [54] implicitly project the spectral data into a high-dimensional kernel Hilbert space, where linear algorithms can approximate nonlinear relationships. Common kernels include the Gaussian, polynomial, and radial basis function (RBF) [55], [56] kernels, often combined with support vector regression or kernel-based least squares unmixing. Finally, manifold learning-based unmixing techniques [57], [58], [59] have been developed to exploit the intrinsic geometry of hyperspectral data. These approaches are grounded in the assumption that the nonlinear dataset lies on a smooth, low-dimensional manifold embedded within the high-dimensional spectral space. By leveraging tools such as Laplacian regularization and locally linear embedding, these methods aim to preserve geometric consistency and improve unmixing accuracy in complex nonlinear scenarios.

B. Deep-Learning-Based Approaches and Their Limitations

In recent years, deep learning [60], [61] has been increasingly applied to hyperspectral unmixing due to its powerful capabilities in feature extraction and nonlinear modeling. However, most existing deep-learning-based unmixing meth-

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ods [62], [63], [64] are still based on the LMM, where the primary objective is to estimate abundances and endmember spectra under the assumption of linearity. These methods typically adopt supervised or unsupervised neural network architectures, e.g., autoencoders [65], convolutional neural networks (CNNs) [66], [67], transformer [68], or recurrent neural networks (RNNs) [69], to learn mappings from pixel spectra/image patches to abundance vectors, often incorporating constraints such as nonnegativity and sum-to-one during training. Representative examples include the abundance-guided spectral and spatial network (A2SN) [70], multistage convolutional autoencoder network (MSNet) [71], dual-feature fusion network [72], and minimum simplex convolutional network (MiSiCNet) [73]. However, despite these advances, they still share the fundamental limitations of the linear model.

To address this, researchers have begun exploring deep-learning-based nonlinear unmixing strategies that move beyond the LMM framework [74], [75]. These methods aim to leverage the nonlinear modeling capacity of neural networks, typically by implicitly learning the mixing process from the data. A few studies have also explored explicitly incorporating nonlinear physical priors into the network architecture. For instance, Wang et al. [76] developed a novel blind nonlinear autoencoder unmixing network (NLAEU). The encoder and decoder are specifically designed to enable the simultaneous extraction of both endmembers and abundances from hyperspectral data. Chen et al. [77] presented a spatial–spectral adaptive nonlinear unmixing network (SSANU-Net) that effectively captures spatial–spectral features through a dual-stream encoder architecture while accurately modeling both linear and nonlinear photon scattering phenomena via a corresponding two-stream decoder. Zhao et al. [78] proposed a novel nonlinear unmixing method based on a long short-term memory network and a deep neural network (LSTM–DNN) to effectively characterize the spectral sequential properties of hyperspectral pixels. The proposed LSTM–DNN autoencoder architecture simultaneously exploits spectral band correlations, interband variability, and spatial information during network training while incorporating total variation regularization to enforce spatial smoothness in the abundance maps.

While recent deep-learning-based nonlinear unmixing methods have achieved promising results, they fail when pure pixels are absent from the dataset. In such cases, virtual endmembers must be estimated by identifying the smallest simplex that encloses the data. However, the minimum simplex volume constraint (MSVC), originally developed for linear scenarios, is ineffective under these conditions. This limitation restricts the use of traditional endmember regularization strategies and often results in suboptimal estimation of both abundances and endmembers. In addition, the majority of existing nonlinear unmixing methods still rely on increasing network depth or applying nonlinear activation functions, rather than systematically modeling the underlying physical interactions between light and materials.

To overcome these limitations, we propose MPGU-Net, a novel end-to-end multimodel physics-guided unmixing network for nonlinear unmixing in no-pure-pixel scenarios. Our method contains three main innovations as follows.

1) MPGU-Net explicitly integrates nonlinear physical models into the deep learning framework to characterize light–material interactions and achieve generalizability across multiple mixing scenarios. We embed two representative nonlinear mixing models, i.e., the PPNM and the MLM, into the reconstruction module to explicitly capture a broad spectrum of nonlinear interactions. Unlike conventional approaches that depend on implicit nonlinear representations through additional network layers, our method directly reconstructs hyperspectral signatures using physically meaningful nonlinear formulations, yielding both more accurate and interpretable unmixing results across diverse nonlinear scenarios.
2) MPGU-Net addresses nonpure pixel scenarios with a novel linearizing space transformation (LST) that analytically approximates the linear component of a nonlinear dataset. This crucial innovation allows for the use of geometric constraints (particularly the minimum simplex volume criterion) for endmember regularization in nonlinear no-pure-pixel mixing environments.
3) MPGU-Net employs an integrated loss formulation that combines reconstruction fidelity, geometric regularization on the linear component of the nonlinear dataset estimated by LST, directional coherence of endmembers, and latent regularization. Its contribution lies in how these commonly used loss components are systematically organized and coupled with the proposed LST and physics-guided reconstruction models, enabling consistent optimization under nonlinear and no-pure-pixel scenarios.

With these three innovations, the proposed framework exhibits strong potential for accurately unmixing complex nonlinear datasets where pure pixels are scarce or absent. This significantly expands its practical applicability for real-world hyperspectral datasets.

C. Article Organization

The remaining sections of this article are organized as follows. Section II briefly describes some preliminaries. Section III details the network structure of the proposed MPGU-Net. Section IV presents extensive experiments and analyses on synthetic and real unmixing datasets. Section V further discusses the behavior of the proposed method under different conditions. Section VI concludes this article.

II. Preliminaries: Hyperspectral Mixing Models

Since the proposed method generalizes to multiple mixing scenarios, this section provides a brief introduction to the relevant concepts and background of hyperspectral unmixing. The spectral reflectances of the hyperspectral dataset $\mathbf{Y}(\{\mathbf{y}_n\}_{n=1}^{N_p} \in \mathbf{R}_+^{N_b})$ composed of N_b spectral bands and N_p samples can be modeled as

$$\mathbf{y}_n = F(\mathbf{M}\mathbf{a}_n) + \boldsymbol{e}_n \tag{1}$$

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where $\mathbf{M} \in \mathbb{R}^{N_b \times c}$ represents the endmember matrix containing c pure spectral signatures, $\mathbf{a}_n \in \mathbb{R}^{c \times 1}$ corresponds to the abundance vector, $\mathbf{e}_n \in \mathbb{R}^{N_b \times 1}$ accounts for additive noise for the nth pixel, and F is a nonlinear function. By varying the choice of F, different mixing models can be derived.

A. Linear Mixing Model

The LMM serves as a fundamental mathematical framework in hyperspectral and multispectral remote sensing for analyzing mixed pixels that contain spectral contributions from multiple distinct materials (endmembers). The model operates on the core assumption that the observed spectral reflectance of any given pixel can be reconstructed by linearly combining endmembers and their corresponding fractional abundances. Mathematically, this relationship can be expressed as

$$\mathbf{y}_n = F(\mathbf{M}\mathbf{a}_n) + \mathbf{e}_n = \mathbf{M}\mathbf{a}_n + \mathbf{e}_n. \tag{2}$$

Since the fractional abundance represents the proportion of ground area covered by each pure material, this model is typically subject to two physical constraints: the abundance nonnegative constraint (ANC), requiring all abundance values to be nonnegative ($a_{ij} \geq 0$), and the abundance sum-to-one constraint (ASC), ensuring the abundances sum to unity for each pixel ($\sum_{i=1}^{c} a_{ij} = 1 \quad \forall j$).

B. Polynomial PPNM

The polynomial PPNM [43] extends the conventional LMM by accounting for nonlinear effects commonly observed in scenes with vegetation canopies. It assumes that the incident ray of light undergoes second-order interactions before reaching the sensor. The PPNM can be mathematically expressed as

$$\mathbf{y}_n = F(\mathbf{M}\mathbf{a}_n) + \mathbf{e}_n = \mathbf{M}\mathbf{a}_n + p_n(\mathbf{M}\mathbf{a}_n) \odot (\mathbf{M}\mathbf{a}_n) + \mathbf{e}_n \tag{3}$$

where p_n is the nonlinear parameter for the nth pixel. The symbol $\odot$ denotes the Hadamard (elementwise) product. The term $p_n(\mathbf{M}\mathbf{a}_n) \odot (\mathbf{M}\mathbf{a}_n)$ captures the second-order nonlinear interactions between endmembers, enabling the model to more accurately represent hyperspectral data from complex scenes. The value of p_n can vary between -1 and 1. Similar to the LMM, the abundance vector usually satisfies the two constraints, i.e., ANC and ASC.

C. Multilinear Mixing Model

The MLM [46] is an advanced nonlinear mixing model that explicitly models multiple scattering phenomena within a pixel. It assumes that the incident ray of light interacts with multiple pure materials before being recorded by the sensor. This leads to an infinite series representation, which can be compactly written as

$$\mathbf{y}_n = F(\mathbf{M}\mathbf{a}_n) + \mathbf{e}_n = \frac{(1 - p_n)\mathbf{M}\mathbf{a}_n}{(1 - p_n\mathbf{M}\mathbf{a}_n)} + \mathbf{e}_n \tag{4}$$

where p_n is a pixel-dependent nonlinear coefficient that quantifies the probability of multiple interactions between photons and pure materials within the nth pixel. A large p_n indicates stronger multiple scattering effects, where $p_n = 0$ corresponds to purely linear mixing.

III. Methodology

A. Linearizing Space Transformation

In this work, we developed a new MPGU-Net to estimate endmembers and fractional abundances in nonlinearly mixed hyperspectral datasets where no pure pixels are available. The key concept for achieving the latter is analytically estimating the linear component of a nonlinear dataset. This allows for the use of geometric constraints, in particular the minimum simplex volume criterion, thereby allowing for a more accurate estimation of virtual endmembers.

The proposed method can accommodate any mixing model consisting of an explicit addition of linear and nonlinear terms. When the mixing equation includes multiple nonlinear functions (e.g., polynomials, exponentials, logarithmic functions, and sine/cosine functions), each component can be approximated by decomposing it into linear and nonlinear terms via a Taylor series expansion, provided that the nonlinear functions are sufficiently smooth and differentiable. As an example, general polynomial mixing models are a class of models that extend (1) via a Taylor series expansion

$$\begin{aligned}
\mathbf{y}_n &= F(\mathbf{x}_n) \\
&= \mathbf{x}_n + p_n\mathbf{x}_n \odot \mathbf{x}_n + p_n^2\mathbf{x}_n \odot \mathbf{x}_n \odot \mathbf{x}_n \\
&\quad + p_n^3\mathbf{x}_n \odot \mathbf{x}_n \odot \mathbf{x}_n \odot \mathbf{x}_n + \cdots \\
&= \mathbf{x}_n + p_n\mathbf{x}_n \odot \Big(\mathbf{x}_n + p_n\mathbf{x}_n \odot \mathbf{x}_n + p_n^2\mathbf{x}_n \odot \mathbf{x}_n \odot \mathbf{x}_n \\
&\quad + p_n^3\mathbf{x}_n \odot \mathbf{x}_n \odot \mathbf{x}_n \odot \mathbf{x}_n + \cdots\Big) \\
&= \mathbf{x}_n + p_n\mathbf{x}_n \odot \mathbf{y}_n \\
&= \frac{\mathbf{x}_n}{1 - p_n\mathbf{x}_n}
\end{aligned} \tag{5}$$

where $\mathbf{x}_n = \mathbf{M}\mathbf{a}_n$ is the linear component of the nonlinear dataset.

To estimate the linear component of a nonlinear dataset, we introduce an analytical approach called the LST. The linear term of the polynomial mixing models can be obtained by rearranging (5)

$$\mathbf{x}_n = \frac{\mathbf{y}_n}{1 + p_n\mathbf{y}_n}. \tag{6}$$

When considering only the first two terms of (5), a bilinear mixing model is obtained: $\mathbf{y}_n = \mathbf{x}_n + p_n\mathbf{x}_n \odot \mathbf{x}_n$, and $\mathbf{x}_n$ can be estimated by solving a quadratic equation

$$\mathbf{x}_n = \frac{-1 + \sqrt{1 + 4\ p_n\mathbf{y}_n}}{2p_n}. \tag{7}$$

To demonstrate the impact of this transformation, Fig. 1(a) shows the principal component analysis (PCA) embeddings of hyperspectral data generated with the PPNM model. The dataset is constructed from six endmembers obtained from the United States Geological Survey (USGS) database. Fractional abundances were generated to ensure that at least three binary mixtures are available for each endmember pair (see Section IV-B1 for a detailed description of this dataset). As noted in [73], the minimum volume simplex constraint can only aid in reconstructing virtual endmembers from a linear dataset when enough data points lie on all edges of the linear

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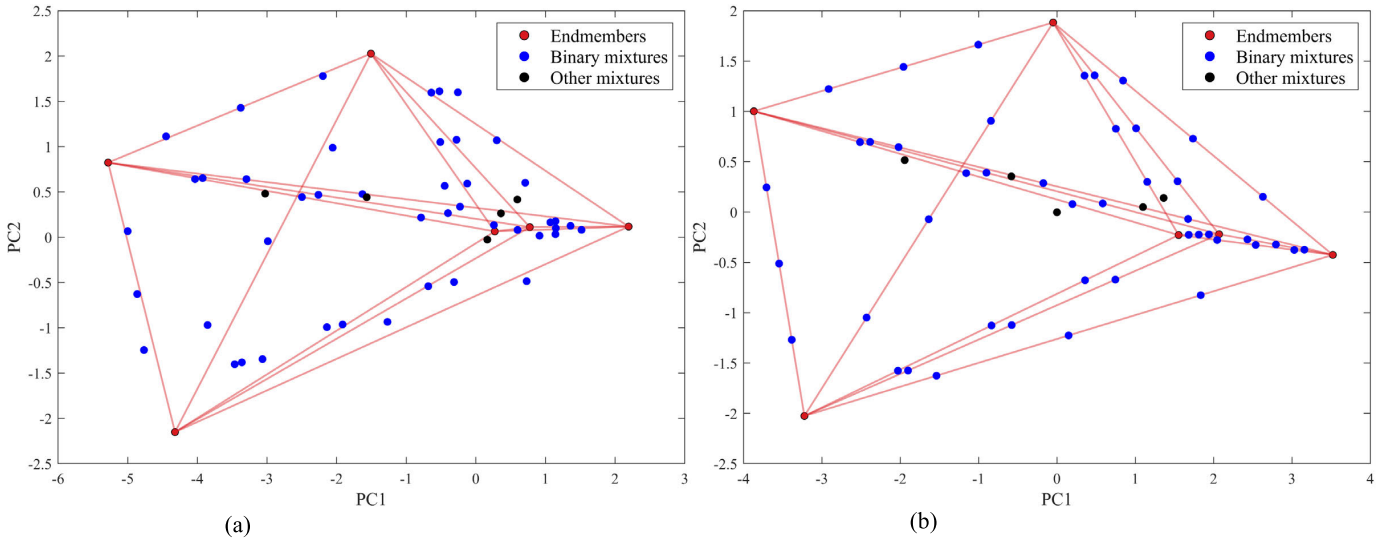


Fig. 1. PCA embeddings of hyperspectral data: (a) under PPNM, many binary mixtures do not lie on the edges of the simplex and (b) after LST, all binary mixtures lie on the edges of the simplex.

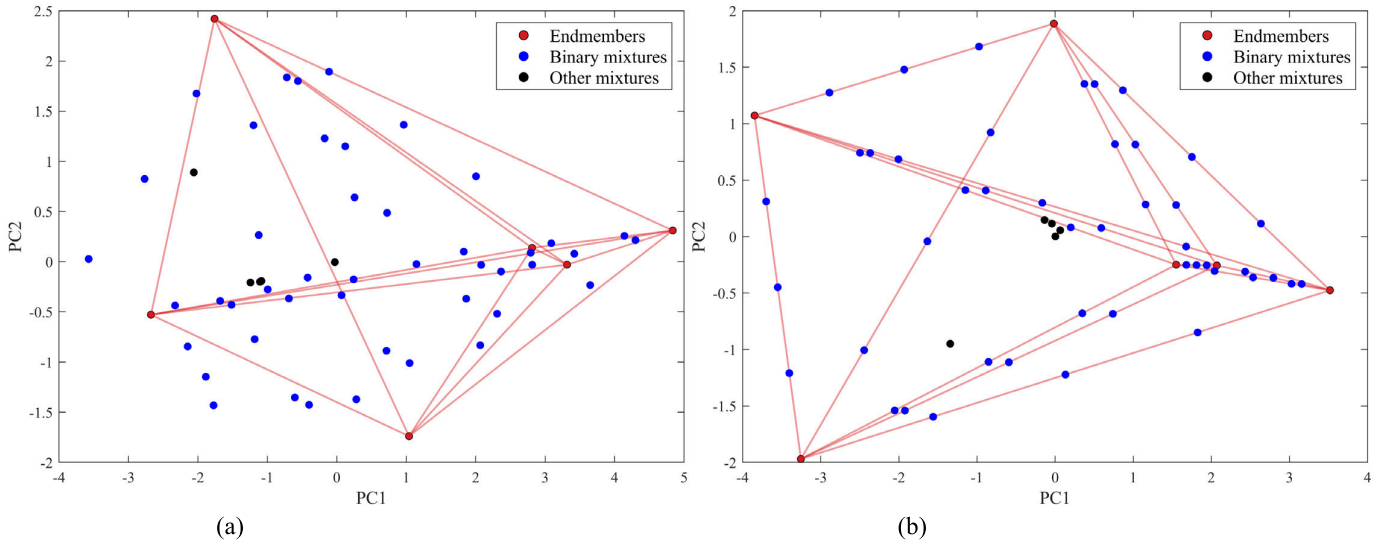


Fig. 2. PCA embeddings of hyperspectral data: (a) under MLM, many binary mixtures do not lie on the edges of the simplex and (b) after LST, all binary mixtures lie on the edges of the simplex.

simplex. Unfortunately, most of the binary mixtures in the nonlinear dataset [see Fig. 1(a)] do not lie on the edges of the linear simplex, making them incompatible with the minimum volume simplex constraint. In contrast, after applying the LST transformation [see Fig. 1(b)], all binary mixtures are mapped into the convex hull and lie exactly on the edges of the linear simplex.

This approach can be easily generalized to a polynomial mixing model of degree n, but it may require a numerical solution to estimate $\mathbf{x}_n$.

Similar to polynomial mixing models, MLM contains both linear and higher order nonlinear terms

$$\begin{aligned}\mathbf{y}_n &= F(\mathbf{x}_n)\\ &= (1-p_n)\,\mathbf{x}_n + (1-p_n)\,p_n\,\mathbf{x}_n \odot \mathbf{x}_n\\ &\quad + (1-p_n)\,p_n^2\,\mathbf{x}_n \odot \mathbf{x}_n \odot \mathbf{x}_n\\ &\quad + (1-p_n)\,p_n^3\,\mathbf{x}_n \odot \mathbf{x}_n \odot \mathbf{x}_n \odot \mathbf{x}_n + \cdots\\ &= (1-p_n)\,\mathbf{x}_n + p_n\mathbf{x}_n \odot \Big((1-p_n)\big(\mathbf{x}_n + p_n\mathbf{x}_n \odot \mathbf{x}_n\\ &\quad + p_n^2\mathbf{x}_n \odot \mathbf{x}_n \odot \mathbf{x}_n + \cdots\big)\Big)\\ &= (1-p_n)\,\mathbf{x}_n + p_n\mathbf{x}_n \odot \mathbf{y}_n. \end{aligned} \tag{8}$$

The linear term of the MLM model can be obtained by rearranging (8)

$$\mathbf{x}_n = \frac{\mathbf{y}_n}{1-p_n+p_n\mathbf{y}_n}.. \tag{9}$$

Fig. 2(a) shows the data manifold of the MLM dataset, while Fig. 2(b) displays the LST-transformed version. Similar to the PPNM dataset, six endmembers were selected from the USGS database, and fractional abundances were generated to ensure that at least three binary mixtures are available for each

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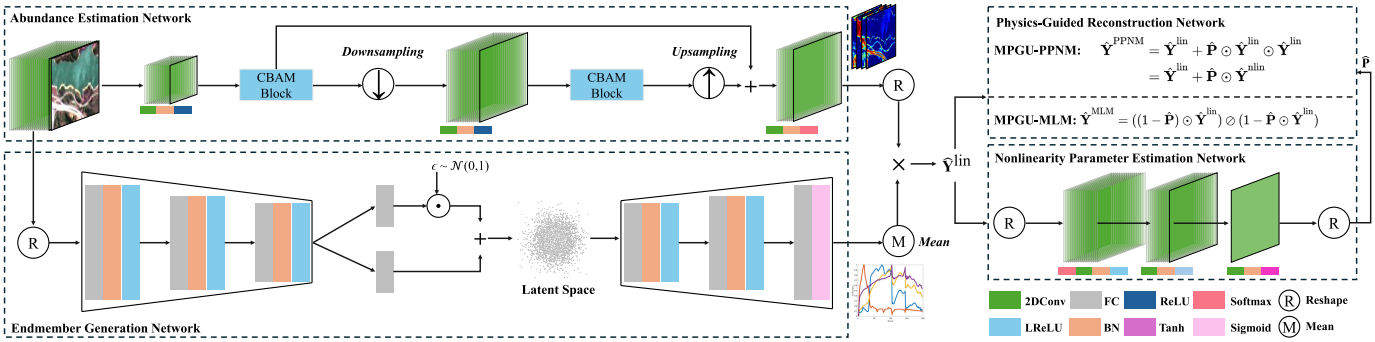


Fig. 3. Architecture of the proposed MPGU-Net, including an endmember generation network, an abundance estimation network, a nonlinearity parameter estimation network, and a physics-guided reconstruction network. In the latter, the two nonlinear models are made explicit, leading to MPGU-PPNM and MPGU-MLM, respectively.

endmember pair. Again, LST maps all binary mixtures onto the edges of the linear simplex.

As mentioned in the introduction, endmember estimation in blind nonlinear unmixing remains particularly challenging in scenarios without pure pixels. Many existing blind nonlinear unmixing methods primarily rely on reconstruction fidelity or implicit nonlinear modeling, and do not explicitly enforce geometric constraints on the endmember simplex. While these methods can achieve accurate results in certain cases, the absence of simplex volume regularization may lead to less stable or less geometrically consistent endmember estimates in nonlinear no-pure-pixel scenarios. The MSVC, designed for no-pure-pixel scenarios, is incompatible with nonlinear scenarios. By appropriately approximating the linear component of a dataset, the proposed LST enables the use of MSVC. Our proposed LST strategy offers three main advantages: 1) it provides a viable approach for applying MSVC under nonlinear conditions; 2) it improves the accuracy of endmember and abundance estimation by aligning the linear component predicted by LST with that estimated by our unmixing network; and 3) it serves as a consistency check between the input spectra and the estimated nonlinear parameters of the mixing models. Note that LST is not a direct part of our network architecture, but performs these three tasks via the loss terms (see Section III-F for detailed information).

Because the proposed MPGU-Net must simultaneously estimate endmembers, fractional abundances, and mixing-model parameters, while at the same time reconstructing the input, four dedicated networks were designed: 1) an endmember generation network; 2) an abundance estimation network; 3) a nonlinearity parameter estimation network; and 4) a physics-guided reconstruction network. The architecture of MPGU-Net is illustrated in Fig. 3.

B. Endmember Generation Network

To improve the robustness of endmember extraction under varying noise, illumination, and atmospheric conditions, we design a probabilistic endmember generation module based on a variational autoencoder (VAE) framework [79]. This module learns a flexible latent representation of global endmember distributions from the observed hyperspectral data, providing an explicit regularization on the endmember space. Although the VAE operates at the pixel level during training, the final endmember matrix used for unmixing is obtained by averaging the generated endmember candidates over all pixels, which guarantees a stable and globally consistent representation for subsequent reconstruction.

1) Generative Model: According to (1), each observed hyperspectral pixel $\mathbf{y}_n$ can be described as a nonlinear mixture of endmember signatures. A probabilistic formulation of a spectrum is given by

$$\mathbf{y}_n \mid (\mathbf{M}, \mathbf{a}_n, p_n) \sim \mathcal{N}\left(F\left(\mathbf{M}\mathbf{a}_n\right), \mathbf{\Lambda}\right) \tag{10}$$

where $\mathbf{\Lambda} = \lambda_0^{-1}\mathbf{I}$ is the covariance matrix of the observation noise, with $\lambda_0 > 0$ (a scalar) denoting the noise precision (inverse of the variance). This probabilistic formulation models the observed spectrum $\mathbf{y}_n$ as a random realization from a Gaussian distribution with a mean corresponding to the ideal nonlinear mixture predicted by a mixing model, and a covariance that accounts for deviations caused by measurement noise, illumination changes, atmospheric effects, and other modeling inaccuracies. This formulation allows for random fluctuation of $\mathbf{y}_n$ around the ideal spectrum.

To obtain the global endmember matrix $\mathbf{M}$, we introduce a latent variable $\mathbf{z}_n$ for each pixel, representing a low-dimensional code that captures the global spectral characteristics of the scene. This latent variable is learned jointly with the unmixing process, and can be interpreted as a compact description of the underlying endmember distribution that produced pixel $\mathbf{y}_n$. The mapping from $\mathbf{z}_n$ to the corresponding endmember matrix candidate $\mathbf{M}_n$ is parameterized by a decoder network $f_\theta(\cdot)$. The output of the decoder is passed through an elementwise sigmoid activation to ensure that all reflectance values lie in the physically valid range [0, 1]

$$\mathbf{M}_n = \text{Sigmoid}\left(f_\theta\left(\mathbf{z}_n\right)\right). \tag{11}$$

During training, latent codes $\mathbf{z}_n$ are sampled from their approximate posteriors, and each sample produces a candidate $\mathbf{M}_n$ through the decoder. These candidates should be interpreted as stochastic realizations of the underlying global endmember distribution, rather than as pixel-specific endmember matrices.

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To obtain a stable global estimate, the final endmember matrix is computed by averaging over all pixels

$$\hat{\mathbf{M}} = \frac{1}{N_p}\sum_{n=1}^{N_p}\mathbf{M}_n. \tag{12}$$

It is worth noting that although the latent variable $\mathbf{z}_n$ and the corresponding candidate endmember matrix $\mathbf{M}_n$ are inferred at the pixel level, they are not intended to represent pixel-specific endmembers. Instead, each $\mathbf{M}_n$ should be interpreted as a stochastic realization sampled from the underlying global endmember distribution learned by the VAE decoder. By averaging the candidate endmember matrices over all pixels as in (12), the final endmember matrix aggregates information from the entire scene, leading to a stable and globally consistent estimate. This formulation is robust to scenarios with sparse abundances, since pixels in which certain endmembers are absent contribute little to their reconstruction, while pixels where those endmembers are present provide informative gradients. As a result, the averaging process does not bias the estimation toward frequently occurring materials and does not degrade performance when some endmembers are missing in many pixels.

2) Inference Model: While the generative model defines how each pixel is generated from the latent code $\mathbf{z}_n$ through the decoder $f_\theta(\cdot)$ and the nonlinear mixing process [see (10)], the inference model addresses the reverse problem: estimating $\mathbf{z}_n$ from $\mathbf{y}_n$. The true posterior distribution $p(\mathbf{z}_n|\mathbf{y}_n)$ is analytically intractable, primarily because $p(\mathbf{z}_n)$ and $p(\mathbf{y}_n)$ are either unknown or computationally expensive to evaluate. To address this, approximate inference methods such as variational inference and Markov chain Monte Carlo (MCMC) are commonly employed. While MCMC is a popular technique in Bayesian inference for sampling from posterior distributions, it faces challenges in determining whether the samples have converged to the target distribution. This often leads to prolonged convergence times, as samplers may become trapped in local high-probability regions, reducing sample diversity. To overcome these challenges, we adopt stochastic gradient variational Bayes (SGVB) [80], a deterministic approximation method for posterior inference. SGVB not only provides a scalable solution for large datasets but also avoids the convergence issues associated with MCMC. Especially, we use a neural network $q_{\boldsymbol{\phi}_\mathbf{z}}$, parameterized by $\boldsymbol{\phi}_\mathbf{z}$, to approximate the true posterior. The approximate posterior is modeled as a multivariate Gaussian distribution

$$q_{\boldsymbol{\phi}_\mathbf{z}}(\mathbf{z}_n|\mathbf{y}_n) = \mathcal{N}\left(\boldsymbol{\mu}_\mathbf{z}(\mathbf{y}_n), \boldsymbol{\Sigma}_\mathbf{z}(\mathbf{y}_n)\right) \tag{13}$$

with mean vector $\boldsymbol{\mu}_\mathbf{z}(\cdot)$ and diagonal covariance matrices $\boldsymbol{\Sigma}_\mathbf{z}(\cdot) = \mathrm{diag}(\sigma_z^2)$. To enable stochastic sampling from the latent space while maintaining end-to-end differentiability, we apply the reparameterization trick, which expresses the sampling process as

$$\mathbf{z}_n = \boldsymbol{\mu}_\mathbf{z}(\mathbf{y}_n) + \epsilon \odot \boldsymbol{\sigma}_\mathbf{z}(\mathbf{y}_n) \tag{14}$$

where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, with $\mathbf{I}$ being the identity matrix.

3) Network Implementation: Both the encoder and decoder are implemented as fully connected (FC) neural networks to effectively model the global spectral structure of the hyperspectral data. The overall input to the network is the hyperspectral image reshaped into a matrix $\mathbf{Y} \in \mathbb{R}^{N_b \times N_p}$. In the encoder, each input spectrum $\mathbf{y}_n \in \mathbb{R}^{N_b \times 1}$ is passed through three sequential FC layers. The first layer maps the spectral dimension to a hidden space of $32 \times c$, which is then reduced to $16 \times c$, and finally to $4 \times c$. Each layer is followed by batch normalization and LeakyReLU activation to improve training stability and nonlinear feature extraction. The encoder outputs the mean vector $\boldsymbol{\mu}$ and the log-variance vector $\log \boldsymbol{\sigma}^2$, which define the approximate posterior distribution over the latent variable $\mathbf{z}_n \in \mathbb{R}^{J \times 1}$. The dimension of $\mathbf{z}_n$ depends on the dataset. In our experiments, we will set J to be between 2 and 4.

The decoder reconstructs the endmember matrix from the latent vector $\mathbf{z}_n$ using three FC layers. The latent vector is first projected to size $4 \times c$, then expanded to $64 \times c$, and finally mapped to $N_b \times c$. Batch normalization and LeakyReLU activations are applied to intermediate layers, and a final sigmoid activation ensures physically valid outputs. By applying this process to all N_p pixel spectra, the decoder generates a set of endmember matrix candidates $\{\mathbf{M}_n\}_{n=1}^{N_p}$, which are stacked into a tensor of shape $\mathbb{R}^{N_p \times N_b \times c}$. The final global endmember matrix $\hat{\mathbf{M}} \in \mathbb{R}^{N_b \times c}$ is then obtained by averaging along the pixel dimension [see (12)]. This global matrix is used in subsequent spectral reconstruction. The latent space is regularized by a Kullback–Leibler (KL) divergence term during training, encouraging a structured and smooth latent distribution.

C. Abundance Estimation Network

The abundance estimation network is designed to estimate abundance maps from the hyperspectral input cube. It adopts a convolutional encoder–decoder architecture enhanced with a convolutional block attention module (CBAM) to refine spectral–spatial feature representations. The input to this module is the hyperspectral image cube $\mathbb{Y} \in \mathbb{R}^{N_b \times W \times H}$, where $W \times H$ represents the spatial dimension. The abundance estimation network begins with a 1×1 convolutional layer that increases the feature dimensionality to 64, followed by batch normalization, ReLU activation, and a CBAM

$$\mathbb{F}_1^{(E)} = \mathrm{CBAM}\left(\mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1,1}\left(\mathbb{Y}\right)\right)\right)\right) \tag{15}$$

where $\mathbb{F}_1^{(E)} \in \mathbb{R}^{64 \times W \times H}$. Next, a 2×2 max-pooling operation is used to reduce the spatial resolution, followed by another convolutional block

$$\mathbb{F}_2^{(E)} = \mathrm{CBAM}\left(\mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1,1}\left(\mathrm{Maxpool}\left(\mathbb{F}_1^{(\mathrm{E})}\right)\right)\right)\right)\right) \tag{16}$$

where $\mathbb{F}_2^{(E)} \in \mathbb{R}^{128 \times W/2 \times H/2}$. To recover the original spatial resolution, the decoder employs a 2×2 transposed convolution

$$\mathbb{F}_1^{(D)} = \mathrm{UpConv}_{2,2}\left(\mathbb{F}_2^{(\mathrm{E})}\right) \tag{17}$$

where $\mathbb{F}_1^{(D)} \in \mathbb{R}^{64 \times W \times H}$. A skip connection is added to fuse high-resolution features from the encoder

$$\mathbb{F}_2^{(D)} = \mathbb{F}_1^{(D)} + \mathbb{F}_1^{(E)}. \tag{18}$$

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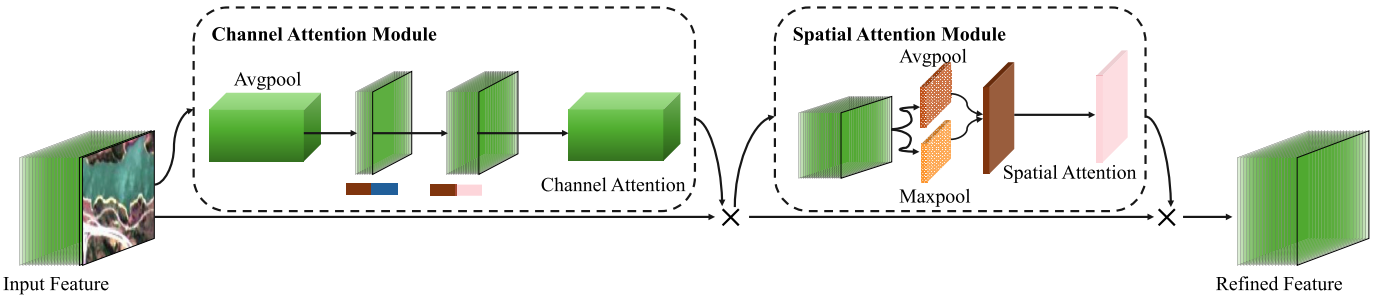


Fig. 4. Architecture of the CBAM, containing a channel attention module and a spatial attention module.

Finally, the output is mapped to abundance maps by a 1×1 convolution, followed by a Softmax across the channel dimension to ensure that the abundance vectors satisfy the two constraints, i.e., ANC and ASC

$$\hat{\mathbb{A}} = \text{Softmax}\left(\text{Conv}_{1,1}\left(\mathbb{F}_2^{(\text{D})}\right)\right) \tag{19}$$

where $\hat{\mathbb{A}} \in \mathbb{R}^{c \times W \times H}$.

1) Convolutional Block Attention Module: To promote more accurate abundance estimation, we adopt a CBAM after each convolutional stage in the abundance estimation network. This design enhances spectral–spatial feature discrimination by allowing the network to focus on informative spectral bands and spatial regions that are most relevant to nonlinear interactions, which cannot be explicitly modeled by standard convolutional operations alone. As a result, the attention mechanism improves robustness against noise and irrelevant features commonly present in hyperspectral data. The architecture of CBAM is illustrated in Fig. 4. Given an intermediate feature map $\mathbb{F} \in \mathbb{R}^{C \times W \times H}$, CBAM first performs global average pooling to encode the spatial statistics of each channel

$$\mathbb{F}_{\text{avg}} = \text{AvgPool}(\mathbb{F}). \tag{20}$$

The pooled feature $\mathbb{F}_{\text{avg}} \in \mathbb{R}^{C \times 1 \times 1}$ is sequentially processed by two 1×1 convolutional layers with a reduction ratio r, followed by a ReLU and a sigmoid activation, to produce the channel attention weights

$$\mathbb{A}_c = \text{Sigmoid}\left(\text{Conv}_{1,1}\left(\text{ReLU}\left(\text{Conv}_{1,1}\left(\mathbb{F}_{\text{avg}}\right)\right)\right)\right) \tag{21}$$

where $\mathbb{A}_c \in \mathbb{R}^{C \times 1 \times 1}$. These weights are applied to the original feature map via the Hadamard (elementwise) product

$$\mathbb{F}_c = \mathbb{F} \odot \mathbb{A}_c. \tag{22}$$

To further capture spatially informative structures, spatial attention is applied to the channel-refined feature $\mathbb{F}_c$. Mean-pooling and max-pooling operations are performed along the channel dimension to obtain two spatial feature maps

$$\begin{aligned} \mathbb{F}_{\text{mean}}(1, W, H) &= \underset{C}{\text{Meanpooling}}\left(\mathbb{F}_c(C, W, H)\right) \\ \mathbb{F}_{\text{max}}(1, W, H) &= \underset{C}{\text{Maxpooling}}\left(\mathbb{F}_c(C, W, H)\right). \end{aligned} \tag{23}$$

These two maps $\mathbb{F}_{\text{mean}}(1, W, H) \in \mathbb{R}^{1 \times W \times H}$ and $\mathbb{F}_{\text{max}}(1, W, H) \in \mathbb{R}^{1 \times W \times H}$ are concatenated to form a two-channel representation and processed by a convolutional layer with a 7×7 kernel and a sigmoid activation

$$\mathbb{A}_s = \text{Sigmoid}\left(\text{Conv}_{7\times 7}\left(\text{Concat}\left(\mathbb{F}_{\text{mean}}, \mathbb{F}_{\text{max}}\right)\right)\right). \tag{24}$$

The final output of the CBAM is obtained by applying the spatial attention weights to $\mathbb{F}_c$

$$\mathbb{F}_{\text{out}} = \mathbb{F}_c \mathbb{A}_s. \tag{25}$$

D. Nonlinearity Parameter Estimation Network

The nonlinearity parameter estimation network is designed to estimate pixelwise nonlinear interaction parameters that characterize deviations from linear mixing. For this purpose, the estimated endmembers and fractional abundances from the previous networks are combined linearly (see Fig. 3). The linearly reconstructed image $\hat{\mathbb{Y}}^{\text{lin}} \in \mathbb{R}^{N_b \times W \times H}$ provides a baseline representation of the scene under the LMM assumption. Any deviations from this baseline implicitly contain information about nonlinear effects. Therefore, the goal of this network branch is to predict a scalar nonlinearity parameter map directly from the linearly mixed image. This design allows the network to learn how much each pixel deviates from purely linear mixing, thereby providing a compact measure of nonlinear interaction strength. Intuitively, pixels whose spectra deviate more from the linear baseline are expected to correspond to higher values of the nonlinearity parameter.

To enhance spectral discrimination, the network starts off with a pointwise convolution followed by a Softmax operation along the spectral dimension

$$\mathbb{P}_0 = \text{Softmax}\left(\text{Conv}_{1,1}\left(\mathbb{Y}^{\text{lin}}\right)\right). \tag{26}$$

The softmax-normalized representation $\mathbb{P}_0 \in \mathbb{R}^{N_b \times W \times H}$ serves as a spectral feature enhancement for downstream convolutional processing. Specifically, we apply two additional convolutional layers with LeakyReLU activations

$$\mathbb{P}_i = \text{LeakyReLU}\left(\text{Conv}_{1,1}\left(\mathbb{P}_{i-1}\right)\right), \quad i = 1, 2. \tag{27}$$

Since measured reflectance values are nonnegative and cannot exceed 1, the nonlinear parameters must be bounded such that the reconstructed reflectance values also remain within the physical range [0, 1]. Through empirical evaluation, we found that setting the upper (and correspondingly lower) bounds of the nonlinear parameter to 0.5 or less satisfies this requirement for both the PPNM and MLM models. To produce a physically meaningful nonlinear parameter map $\mathbb{P}_{\text{out}} \in \mathbb{R}^{1 \times W \times H}$, a final 1×1 convolution followed by a Tanh activation is applied, and the output is then clamped to the range $[-0.5, 0.5]$

$$\hat{\mathbb{P}}_{\text{out}} = \text{Clamp}_{[-0.5,0.5]}\left(\text{Tanh}\left(\text{Conv}_{1,1}\left(\mathbb{P}_2\right)\right)\right). \tag{28}$$

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Finally, $\mathbf{P} = \text{Reshape}(\hat{\mathbb{P}}_{\text{out}}) \in \mathbb{R}^{1\times N_p}$ contains the final estimated nonlinearity parameters.

E. Physics-Guided Reconstruction Network

Finally, a physics-guided reconstruction module is designed to integrate explicit nonlinear mixing models. In this work, we incorporate two representative nonlinear mixing models, namely the PPNM and the MLM, for which the proposed LST transformation admits a closed-form solution. These models are selected as representative examples of commonly encountered nonlinear interactions rather than as an exhaustive description of all possible nonlinear mixing mechanisms in real hyperspectral scenes. For mixing models where closed-form solutions are not available, numerical techniques can be employed to perform the required transformations. Both branches take the estimated endmember matrix $\hat{\mathbf{M}}$, abundance maps $\hat{\mathbf{A}}$, and nonlinear parameter map $\hat{\mathbf{P}}$ as input to generate the reconstructed hyperspectral image. Both branches differ in the way they model interactions of light with matter. The final reconstructed hyperspectral images are computed as

$$\begin{aligned}\hat{\mathbf{Y}}^{\text{PPNM}} &= \hat{\mathbf{Y}}^{\text{lin}} + \hat{\mathbf{P}} \odot \hat{\mathbf{Y}}^{\text{lin}} \odot \hat{\mathbf{Y}}^{\text{lin}} \\ \hat{\mathbf{Y}}^{\text{MLM}} &= \left(\left(1 - \hat{\mathbf{P}}\right) \odot \hat{\mathbf{Y}}^{\text{lin}}\right) \oslash \left(1 - \hat{\mathbf{P}} \odot \hat{\mathbf{Y}}^{\text{lin}}\right)\end{aligned} \tag{29}$$

where $\hat{\mathbf{Y}}^{\text{lin}} = \hat{\mathbf{M}}\hat{\mathbf{A}}$ and $\hat{\mathbf{A}} = \text{Reshape}(\hat{\mathbb{A}}) \in \mathbb{R}^{c\times N_p}$. The symbol $\oslash$ denotes the elementwise division. The network based on the PPNM model is referred to as MPGU-PPNM, while MPGU-MLM denotes the network based on the MLM model.

F. Objective Function

To train our MPGU-Net in an end-to-end manner, we design a multiterm objective function that reflects the roles of the different network components. Since the network simultaneously estimates endmembers, abundances, and nonlinear parameters, the loss must supervise all three aspects while ensuring physically meaningful solutions. These loss terms operate at different levels and serve complementary roles, and are optimized jointly through a weighted formulation. Especially, the objective incorporates a reconstruction loss, an MSVC, a spectral angle-based endmember consistency (SAEC) loss, and a KL divergence regularization.

The reconstruction loss plays a central role, as it connects all outputs of MPGU-Net to the observed spectra. MPGU-Net performs nonlinear reconstruction (NR) through the physics-guided module, while a linear reconstruction (LR) is obtained by multiplying the estimated endmembers and abundances.

To ensure reconstruction fidelity and consistency between different branches, we define

$$\mathcal{L}_{RE} = \frac{1}{N_p}\sum_{j=1}^{N_p}\left(\arccos\frac{\mathbf{y}_j\cdot\hat{\mathbf{y}}_j}{\|\mathbf{y}_j\|_2\|\hat{\mathbf{y}}_j\|_2} + \arccos\frac{\mathbf{x}_j\cdot\hat{\mathbf{y}}_j^{\text{lin}}}{\|\mathbf{x}_j\|_2\|\hat{\mathbf{y}}_j^{\text{lin}}\|_2}\right) \tag{30}$$

where $\hat{\mathbf{y}}_j$ denotes the NR, $\hat{\mathbf{y}}_j^{\text{lin}}$ is the LR, and $\mathbf{x}_j$ represents a reference linear component from LST.

The first term enforces the NR to match the observed spectra, while the second term encourages consistency between the LR and the reference linear component. This cross-branch consistency helps stabilize training and improves the reliability of the estimated endmembers and abundances. To regularize the endmember generation process and prevent arbitrary expansion of the endmember simplex, we introduce the MSVC loss

$$\mathcal{L}_{MSVC} = \|\hat{\mathbf{M}} - \bar{\mathbf{X}}\mathbf{1}_{\text{c}}^{\text{T}}\|_F \tag{31}$$

where $\hat{\mathbf{M}}$ is the estimated endmember matrix, and $\bar{\mathbf{X}} \in \mathbb{R}^{N_b\times 1}$ is the mean vector of $\mathbf{X}(\{\mathbf{x}_j\}_{j=1}^{N_p} \in \mathbf{R}_+^{N_b})$. The matrix $\mathbf{X}$ is estimated using LST. This constraint encourages all endmembers to be centered around the data mean, reducing the volume of the simplex they span and promoting a compact, identifiable spectral space.

To further enforce coherence among the estimated endmembers, we introduce a spectral angle-based regularization that penalizes large deviations from the mean spectral direction

$$\mathcal{L}_{SAEC} = \frac{1}{c}\sum_{i=1}^{c}\arccos\frac{\hat{\mathbf{m}}_i\cdot\bar{\mathbf{X}}}{\|\hat{\mathbf{m}}_i\|_2\|\bar{\mathbf{X}}\|_2} \tag{32}$$

where $\hat{\mathbf{m}}_i$ denotes the ith estimated endmember. This constraint preserves the directional integrity of endmembers, avoiding spectral drift and improving material discrimination.

Since the endmembers are generated through a VAE-based module, we include a KL divergence loss to regularize the latent space. This term encourages the latent variables to follow a standard normal distribution:

$$\mathcal{L}_{KL} = \frac{1}{N_p}\sum_{n=1}^{N_p}\sum_{k=1}^{J} -\frac{1}{2}\left[\log\boldsymbol{\Sigma}_{\mathbf{z}}^{(k)2} - \boldsymbol{\mu}_{\mathbf{z}}^{(k)2} - \boldsymbol{\Sigma}_{\mathbf{z}}^{(k)2} + 1\right] \tag{33}$$

where J is the dimension of the latent space.

The final objective function is a weighted combination of all the above components

$$\mathcal{L} = \mathcal{L}_{RE} + \beta\mathcal{L}_{MSVC} + \gamma\mathcal{L}_{SAEC} + \lambda\mathcal{L}_{KL} \tag{34}$$

where β, γ, and λ are hyperparameters controlling the relative importance of each term. This composite objective function integrates spectral reconstruction accuracy, geometric compactness, angular consistency, and latent space regularization, enabling MPGU-Net to learn accurate and physically interpretable unmixing results across both linear and nonlinear mixture scenarios. As the loss function of the proposed method includes several inverse cosine terms, it is often challenging to optimize due to nonlinearity. In hyperspectral unmixing, however, both predicted and reference reflectance spectra are nonnegative and bounded within [0, 1], which constrains the cosine similarity to the range [0, 1]. As a result, the argument of the arccosine function remains in a well-behaved region during training. The gradient of the arccosine function becomes very large only when the cosine similarity approaches 1, i.e., when the two spectra are nearly parallel. To prevent numerical instability in this extreme case, we add a small positive constant to the denominator before computing the inverse cosine loss. The major advantage of utilizing the inverse cosine loss is that this loss term is less sensitive to scaling effects caused by variations in the illumination and acquisition conditions (incident and measured zenith angles), which are commonly observed in real hyperspectral data and are one of the manifestations of endmember variability.

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By incorporating multiple regularization terms for endmembers, the proposed method can also reliably estimate virtual endmembers from datasets that do not contain pure pixels. These regularization terms are advantageous in both linear and nonlinear scenarios. Because both PPNM and MLM reduce to the linear model only when the estimated nonlinear parameter map $\hat{\mathbf{P}}$ is zero, the closer the estimated $\hat{\mathbf{P}}$ is to zero, the more accurate the resulting abundance and endmember estimates are for linearly mixed datasets.

IV. Experiments

A. Comparative Methods

To comprehensively evaluate the effectiveness of the proposed MPGU-PPNM and MPGU-MLM networks, we compare them with a set of representative linear and nonlinear unmixing methods. Although both MPGU-PPNM and MPGU-MLM are designed as nonlinear unmixing networks, their hyperparameters can be configured to allow them to also operate effectively under the linear mixing assumption, allowing a fair comparison with linear methods. Specifically, two LMM-based unmixing methods, the MSNet [71] and the untied denoising autoencoder with sparsity (uDAS) [81], are included to compare the performance in linear scenarios. In addition, five state-of-the-art nonlinear unmixing methods are selected to benchmark performance in nonlinear scenarios: the two original PPNM [43] and MLM [46] models, a low-rank nonnegative tensor factorization (LR-NTF) method [33], a model-based autoencoder for hyperspectral unmixing (MACU) [82], and a deep learning NLAEU [76]. This setup allows us to assess the flexibility and robustness of the proposed networks under both linear and nonlinear mixing conditions. Note that MLM, PPNM, and LR-NTF employ VCA to initialize the endmember matrix before abundance estimation, so these methods generate identical endmember estimates.

B. Datasets

1) Synthetic Nonlinear Datasets: Two synthetic hyperspectral datasets were generated to evaluate unmixing algorithms under varying degrees of spectral nonlinearity and noise. To avoid bias toward geometric endmember extraction, the datasets were constructed without pure pixels. Each dataset contains 105×105 pixels with 224 spectral bands covering the 400–2500 nm range, synthesized from the same six endmembers but using different mixing mechanisms to control the nonlinearity level. The datasets consist of 49 homogeneous squares, each measuring 5×5 pixels. Of these, 45 squares represent binary mixtures (for each endmember pair, three binary mixtures (0.75/0.25, 0.5/0.5, and 0.25/0.75) were considered), while the remaining four correspond to randomly generated higher order mixtures. All background pixels share identical fractional abundances, given by $1/6 \times [1, 1, 1, 1, 1, 1]^T$. Although this dataset is relatively simple, it is particularly useful for clearly illustrating the quality of the estimated abundance maps. For each mixing model, four noise levels were considered by adding zero-mean Gaussian noise corresponding to SNRs of 20, 30, 40, and 50 dB.

1) *PPNM-Based Synthetic Dataset*: Generated using the PPNM, in which a quadratic term is added to the linear mixture. This formulation introduces second-order distortions of spectral signatures, which alter the shape of the mixed spectra. The resulting nonlinearities are typically noticeable but remain bounded to quadratic effects.
2) *MLM-Based Synthetic Dataset*: Simulated using the MLM, in which multiple scattering is modeled through recursive multiplicative interactions. As a result, the spectral distortions may become stronger than those in PPNM, especially when the nonlinearity parameter is large. This dataset, therefore, reflects complex nonlinear mixing scenarios where endmember separation is significantly reduced in PCA space.

2) Ray Tracing Dataset: The Ray Tracing dataset is a realistic nonlinear dataset that was generated using a fully calibrated virtual citrus orchard ray tracer [83], [84], providing a controlled hyperspectral imaging environment. The dataset contains 20×20 pixels with a spatial resolution of 2 m. Each pixel records reflectance values for 216 spectral bands spanning the wavelength range of 350–2500 nm. To mitigate the effects of water absorption, 31 bands in the ranges 1330–1440 nm and 1770–1950 nm were removed, resulting in 185 bands for analysis. The scene consists of three materials, i.e., soil, weed, and tree. Ground-truth fractional abundance values are available for each hyperspectral pixel. This dataset is available online.[1]

3) Jasper Ridge Dataset: The Jasper Ridge dataset, acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), consists of 521×614 pixels with 224 spectral bands spanning the wavelength range of 380–2500 nm. In this experiment, a 100×100 pixel subimage was extracted with an offset at coordinates (105, 269) to ensure computational efficiency. To reduce the influence of atmospheric effects and water absorption, bands 1–3, 108–112, 154–166, and 220–224 were discarded, resulting in 198 spectral bands for analysis. This subset contains four materials: tree, water, soil, and road. The ground-truth endmembers and abundances are generated under the assumption that the Jasper Ridge dataset is linearly mixed. It will be applied in our experiment to assess the performance of the proposed approach in linear scenarios. The dataset and ground truth for endmembers and abundances are available online.[2]

4) Washington DC Mall Dataset: The Washington DC Mall dataset is an airborne hyperspectral image acquired by the HYDICE sensor over the Washington DC Mall area. The cropped image used in this article contains 319×292 pixels with 191 spectral bands covering the wavelength range from 0.4 to 2.4 μm. Ground-truth data for this scene are publicly available and include multiple land-cover classes, such as grass, tree, road, water, and trail. These materials are commonly used as reference endmembers in hyperspectral unmixing studies.[3] Compared to simpler datasets, the

[1] https://www.biw.kuleuven.be/nonlinear-mixing/raytracer.html
[2] https://rslab.ut.ac.ir/data
[3] https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html

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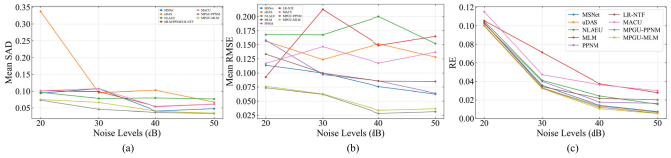


Fig. 5. Performance comparison of various unmixing algorithms on the PPNM-based synthetic dataset at different noise levels: (a) Mean SAD, (b) Mean RMSE, and (c) RE. Note that MLM, PPNM, and LR-NTF share identical VCA-based initialization, leading to identical endmember estimates and thus overlapping curves (identical Mean SAD values).

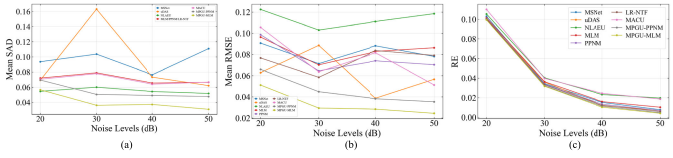


Fig. 6. Performance comparison of various unmixing algorithms on the MLM-based synthetic dataset at different noise levels: (a) Mean SAD, (b) Mean RMSE, and (c) RE. Note that MLM, PPNM, and LR-NTF share identical VCA-based initialization, leading to identical endmember estimates and thus overlapping curves (identical Mean SAD values).

Washington DC Mall scene exhibits more complex spatial structures and material distributions, making it suitable for evaluating the performance of unmixing algorithms in realistic scenarios.

C. Evaluation Metrics

In experiments, we employ three widely used metrics to evaluate the performance of hyperspectral unmixing: spectral angle distance (SAD), abundance root-mean-square error (RMSE), and reconstruction error (RE). These metrics comprehensively assess different aspects of unmixing accuracy with respect to endmember and abundance estimation and image reconstruction.

SAD measures the angular difference between each estimated endmember $\hat{\mathbf{m}}_k$ and its reference $\mathbf{m}_k$ and it is computed as

$$\mathrm{SAD}\left(\mathbf{m}_k, \hat{\mathbf{m}}_k\right) = \arccos \frac{\mathbf{m}_k \cdot \hat{\mathbf{m}}_k}{\|\mathbf{m}_k\|_2 \|\hat{\mathbf{m}}_k\|_2} \tag{35}$$

where a smaller value indicates a higher spectral similarity between the estimated and reference endmembers.

The abundance RMSE is employed to evaluate the abundance estimation accuracy and gives the difference between the estimated abundance $\hat{\mathbf{a}}_j$ and the ground-truth abundance $\mathbf{a}_j$

$$\mathrm{RMSE}\left(\hat{\mathbf{a}}_j, \mathbf{a}_j\right) = \sqrt{\frac{1}{N_p} \sum_{j=1}^{N_p} \left(\hat{\mathbf{a}}_j - \mathbf{a}_j\right)^2}. \tag{36}$$

Finally, RE is used to measure the image reconstruction performance and gives the difference between the reconstructed hyperspectral image and the reference image

$$\begin{aligned} \mathrm{RE}\left(\mathbf{Y}, \hat{\mathbf{Y}}\right) &= \frac{1}{N_p} \sum_{j=1}^{N_p} \mathrm{RE}\left(\mathbf{y}_j, \hat{\mathbf{y}}_j\right) \\ &= \frac{1}{N_p} \sum_{j=1}^{N_p} \arccos \frac{\mathbf{y}_j \cdot \hat{\mathbf{y}}_j}{\|\mathbf{y}_j\|_2 \|\hat{\mathbf{y}}_j\|_2} \end{aligned} \tag{37}$$

D. Experiments on Synthetic Datasets

1) Results on the PPNM-Based Synthetic Dataset: Fig. 5 shows the performance of the compared unmixing algorithms on the PPNM-based synthetic dataset at different noise levels. In Fig. 5(a), the proposed MPGU-PPNM and MPGU-MLM consistently achieve the lowest or near-lowest Mean SAD values across all SNR levels, indicating more accurate endmember estimation. Their advantage becomes more evident as the noise level decreases. At higher SNR levels (40 and 50 dB), both methods achieve Mean SAD values around 0.03–0.04, outperforming most competing approaches. Fig. 5(b) reports the Mean RMSE of abundance estimation. The proposed MPGU-based methods exhibit competitive and stable performance across different noise levels. In particular, MPGU-MLM achieves the lowest RMSE at 40 and 50 dB, with values around 0.03–0.04, while MPGU-PPNM also maintains relatively low estimation errors. Compared with several baseline methods whose performance fluctuates significantly across SNR levels, the proposed methods demonstrate better robustness under varying noise conditions. Fig. 5(c) illustrates the RE for different methods. As expected, the RE decreases as the noise level decreases for all approaches. Nevertheless, MPGU-PPNM and MPGU-MLM consistently achieve among the lowest REs across all SNR levels. At 50 dB, their RE values are close to 0.01, confirming their strong ability to preserve spectral fidelity in the reconstructed signals. Overall, these results demonstrate that the proposed MPGU-based methods provide accurate and robust performance across different noise conditions, particularly in low-noise scenarios where both endmember estimation and abundance estimation are further improved. We also performed experiments by varying the maximum purity level from 0.6 to 0.7, and the results are not significantly different.

2) Results on the MLM-Based Synthetic Dataset: Fig. 6 illustrates the performance of different unmixing methods on the MLM-based synthetic dataset under varying noise levels. As observed in Fig. 6(a), the proposed MPGU-PPNM and MPGU-MLM exhibit consistently low Mean SAD values across different SNRs. In particular, MPGU-MLM achieves the smallest SAD at high SNR levels, approaching 0.03 at 50 dB, while most baseline methods remain above 0.05. This indicates that the proposed framework can more effectively capture nonlinear spectral variations. The abundance estimation results in Fig. 6(b) further demonstrate the advantage of the proposed methods. MPGU-MLM maintains the lowest RMSE across all noise levels, with values remaining around 0.02–0.03 under moderate and high SNR conditions. Meanwhile, MPGU-PPNM also achieves competitive performance with relatively low estimation errors. In contrast, several baseline methods show noticeable fluctuations as the noise level changes, reflecting limited stability in nonlinear scenarios. Fig. 6(c) presents the reconstruction performance in terms of RE. As the SNR increases, all methods exhibit decreasing REs. Nevertheless, the proposed methods consistently achieve the smallest or near-smallest RE values. At 50 dB, the RE of MPGU-PPNM and MPGU-MLM drops to approximately

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TABLE I

PERFORMANCE EVALUATION OF THE DIFFERENT UNMIXING METHODS ON THE MLM-BASED SYNTHETIC DATASET WITH 30-DB NOISE. BEST RESULTS ARE SHOWN IN BOLD, AND SECOND-BEST RESULTS ARE UNDERLINED. NOTE THAT MLM, PPNM, AND LR-NTF SHARE IDENTICAL VCA-BASED INITIALIZATION, LEADING TO IDENTICAL ENDMEMBER ESTIMATES

Methods		MSNet	uDAS	NLAEU	MLM	PPNM	LR-NTF	MACU	MPGU-PPNM	MPGU-MLM
SAD	End 1	0.1205	0.2352	0.1685	0.1840	0.1840	0.1840	0.1838	0.1195	**0.1028**
	End 2	0.1062	0.0290	0.0512	0.0888	0.0888	0.0888	0.0880	0.0915	**0.0150**
	End 3	0.1279	0.2306	0.0270	0.0884	0.0884	0.0884	0.0888	0.0455	**0.0417**
	End 4	0.0617	0.1420	0.0262	0.0230	0.0230	0.0230	0.0233	**0.0110**	0.0275
	End 5	0.1244	0.0883	0.0754	0.0785	0.0785	0.0785	0.0778	0.0254	**0.0081**
	End 6	0.0813	0.2531	0.0119	0.0115	0.0115	0.0115	0.0115	**0.0100**	0.0214
Mean SAD		0.1037	0.1630	0.0600	0.0790	0.0790	0.0790	0.0789	0.0505	**0.0361**
RMSE	End 1	0.0698	0.1223	0.0301	0.0387	0.0554	0.0427	0.0537	**0.0210**	0.0246
	End 2	0.1276	0.0685	0.1589	0.0976	0.0676	0.0724	0.0540	0.0659	**0.0311**
	End 3	0.0880	0.1605	0.0738	0.0925	0.1264	0.1263	0.1136	0.0836	**0.0429**
	End 4	0.0441	0.0705	0.1047	0.0754	0.0339	0.0327	0.0486	0.0389	**0.0232**
	End 5	0.0540	0.0284	0.1672	0.0635	0.0409	0.0378	0.0467	0.0355	**0.0272**
	End 6	0.0444	0.0811	0.0830	0.0533	0.0626	0.0388	0.0648	**0.0243**	0.0275
Mean RMSE		0.0713	0.0886	0.1030	0.0702	0.0645	0.0585	0.0635	0.0449	**0.0294**
RE		0.0344	0.0359	0.0403	0.0364	0.0338	0.0332	0.0397	0.0321	**0.0317**

0.005–0.01, indicating accurate reconstruction of hyperspectral signals under nonlinear mixing conditions. Overall, the results demonstrate that the proposed MPGU-based framework achieves reliable and stable performance across varying noise levels, with particularly strong advantages in high-quality observation conditions.

To further evaluate the performance of the compared methods, Table I reports the quantitative performance of different unmixing methods on the MLM-based synthetic dataset with 30-dB noise in terms of SAD, RMSE, and RE. The proposed MPGU-PPNM and MPGU-MLM achieve the best overall performance among all compared methods. In terms of endmember estimation, MPGU-MLM attains the lowest Mean SAD value of 0.0361, followed by MPGU-PPNM with 0.0505, both significantly outperforming all competing approaches. For most individual endmembers, the proposed methods also achieve the lowest or second-lowest SAD values, demonstrating consistently accurate spectral reconstruction. For abundance estimation, the superiority of the proposed framework is further confirmed by the RMSE results. MPGU-MLM achieves the lowest Mean RMSE of 0.0294, while MPGU-PPNM ranks second with 0.0449, outperforming most baseline methods by a clear margin. Notably, for several endmembers, the RMSE values of the proposed methods are substantially lower than those of competing approaches, indicating improved abundance estimation under nonlinear mixing conditions. In addition, the proposed methods also achieve the lowest RE values, confirming their strong reconstruction capability.

Fig. 7 presents the estimated endmember spectra on the MLM-based synthetic dataset. Several baseline methods exhibit noticeable spectral distortions in multiple bands, especially for endmembers 2 and 3, indicating limited robustness to nonlinear mixing effects. Although traditional nonlinear models can capture the general spectral trends, they fail to accurately reproduce fine spectral details. In contrast, MPGU-PPNM and MPGU-MLM consistently produce spectral curves that closely match the ground truth, with their curves largely overlapping across most wavelengths and preserving both global structures and subtle spectral variations.

Fig. 8 shows the abundance maps estimated by different methods. Several baselines suffer from severe noise, spatial distortions, or blurred material boundaries, particularly in highly mixed regions. In contrast, MPGU-PPNM and MPGU-MLM generate abundance maps that closely resemble the ground truth, with clearer material boundaries and significantly reduced noise. These results demonstrate the superior spatial accuracy and robustness of the proposed framework under strong nonlinear mixing conditions.

E. Experiments on Real Datasets

1) Results on the Ray Tracing Dataset: Table II reports the quantitative unmixing performance of the compared methods on the Ray Tracing dataset. In terms of endmember estimation accuracy, the SAD results reveal noticeable class-dependent variations for most baseline methods. For example, MSNet and uDAS exhibit relatively large errors on the soil and weed classes, while some traditional nonlinear models achieve lower SAD values on specific classes but lack consistency across different materials. In contrast, MPGU-PPNM and MPGU-MLM maintain relatively low and stable SAD values across

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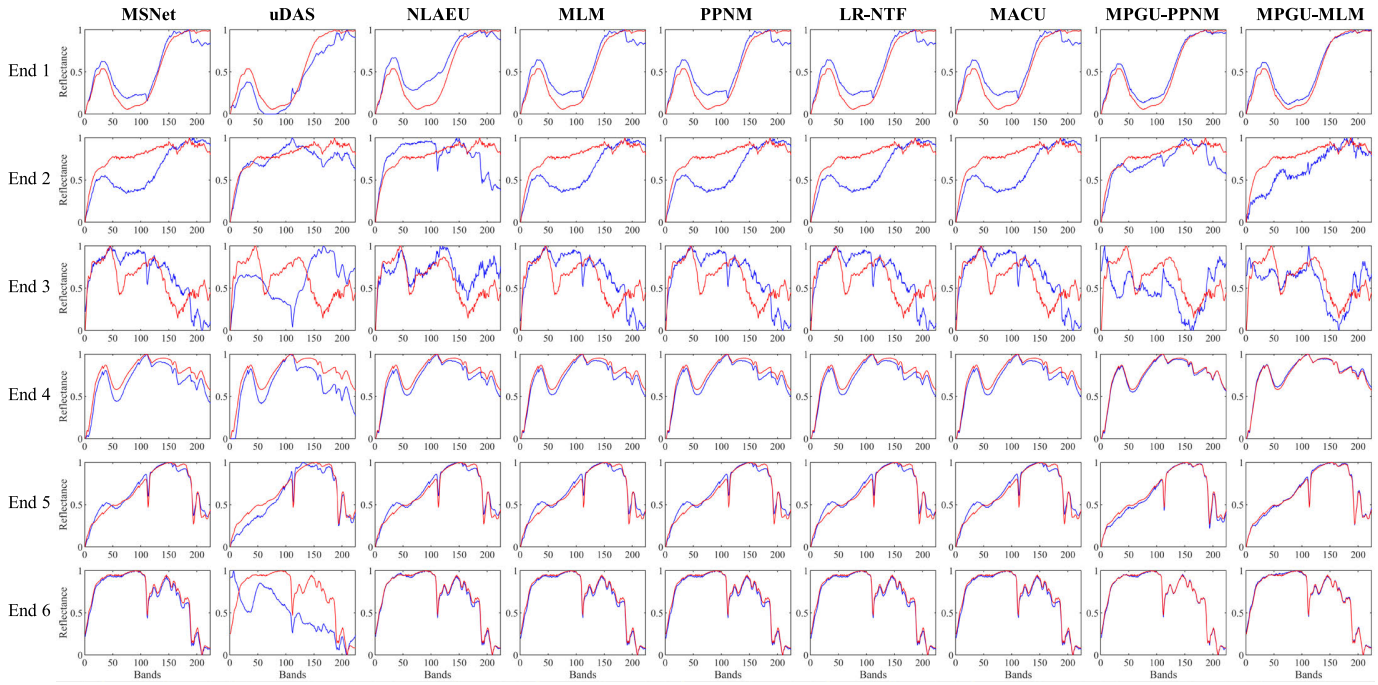


Fig. 7. Visual comparison of the endmembers obtained by different unmixing methods on the MLM-based synthetic dataset with 30-dB noise. Blue curves denote the estimated endmembers, and the red curves denote the ground-truth endmembers. Note that MLM, PPNM, and LR-NTF share identical VCA-based initialization, leading to identical endmember estimates.

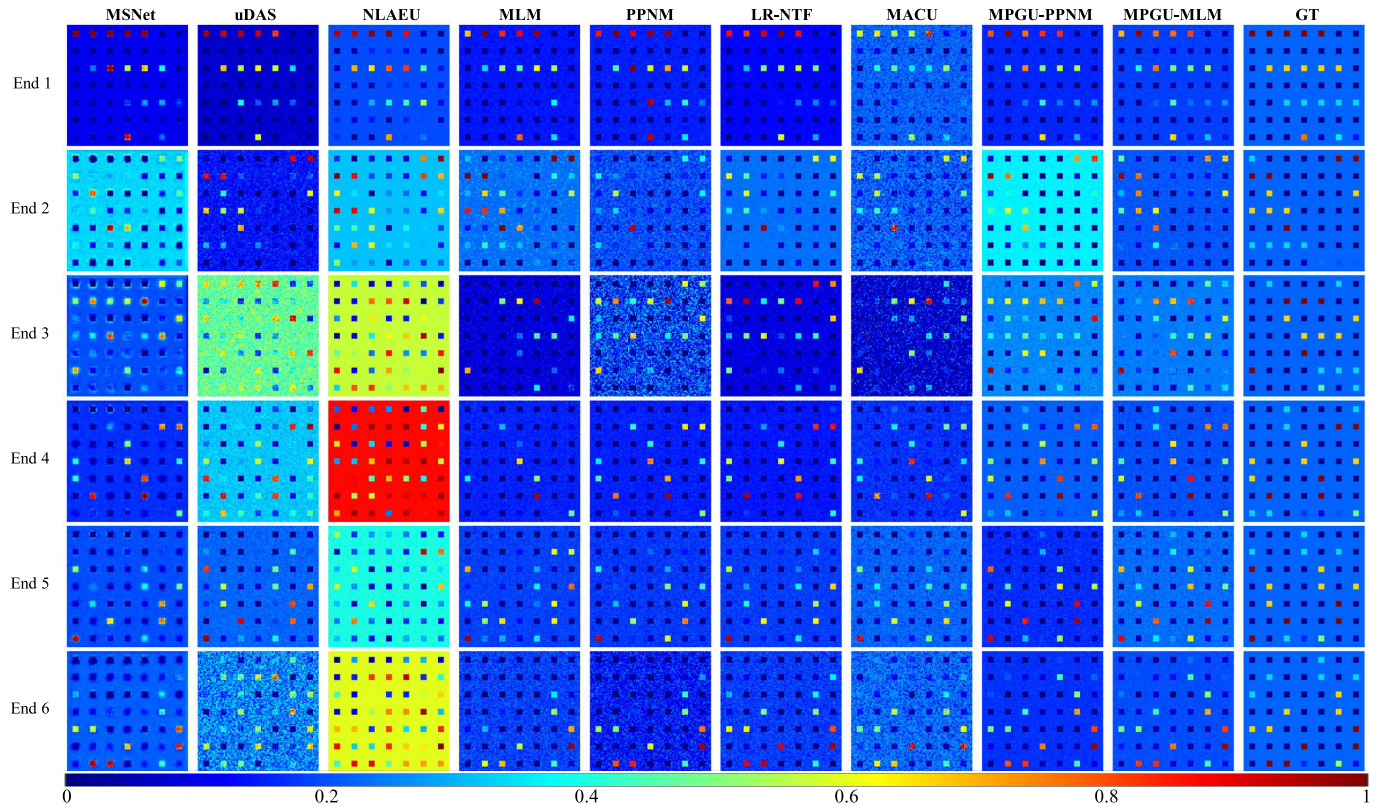


Fig. 8. Abundance maps from the different hyperspectral unmixing methods on the MLM-based synthetic dataset with 30-dB noise. End 1, End 2, End 3, End 4, End 5, and End 6 correspond to the six distinct materials. GT denotes the ground-truth abundances.

all three classes, achieving the lowest and second-lowest Mean SAD, which indicates more robust and reliable endmember extraction under nonlinear mixing conditions. For abundance estimation, the RMSE results further highlight the limitations of baseline approaches in complex mixtures. Several methods suffer from significant error increases on the Weed class,

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TABLE II

PERFORMANCE EVALUATION OF THE DIFFERENT UNMIXING METHODS ON THE RAY TRACING DATASET. BEST RESULTS ARE SHOWN IN BOLD, AND SECOND-BEST RESULTS ARE UNDERLINED. NOTE THAT MLM, PPNM, AND LR-NTF SHARE IDENTICAL VCA-BASED INITIALIZATION, LEADING TO IDENTICAL ENDMEMBER ESTIMATES

Methods		MSNet	uDAS	NLAEU	MLM	PPNM	LR-NTF	MACU	MPGU-PPNM	MPGU-MLM
SAD	Soil	0.1516	0.1502	0.1399	0.1403	0.1403	0.1403	0.1426	**0.0888**	0.0922
	Weed	0.1476	0.2242	0.1688	0.1690	0.1690	0.1690	0.1621	0.1147	**0.1134**
	Tree	**0.0334**	0.0518	0.0622	0.0618	0.0618	0.0618	0.0770	0.0680	0.0729
Mean SAD		0.1109	0.1421	0.1236	0.1237	0.1237	0.1237	0.1272	**0.0905**	0.0928
RMSE	Soil	0.2011	**0.0603**	0.1137	0.2172	0.2472	0.2265	0.2140	0.1196	0.1201
	Weed	0.3224	**0.1521**	0.2572	0.3723	0.3674	0.3204	0.3873	0.1795	0.1930
	Tree	0.1811	0.1838	0.1959	0.1945	0.1578	0.2288	0.2120	**0.0989**	0.1041
Mean RMSE		0.2349	**0.1321**	0.1889	0.2614	0.2575	0.2586	0.2711	0.1327	0.1391
RE		0.0194	0.0225	0.0218	0.0214	0.0231	0.0123	0.0204	0.0146	**0.0119**

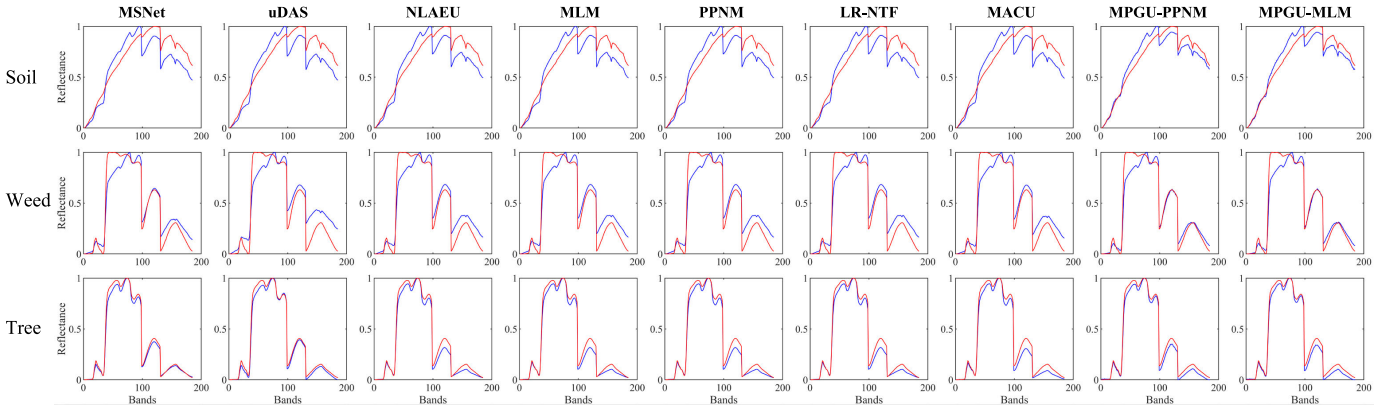


Fig. 9. Visual comparison of the endmembers obtained by different unmixing methods on the Ray Tracing dataset. Blue curves denote the estimated endmembers, and the red curves denote the ground-truth endmembers. Note that MLM, PPNM, and LR-NTF share identical VCA-based initialization, leading to identical endmember estimates.

where RMSE values are considerably higher compared to other materials. In contrast, MPGU-PPNM consistently yields lower RMSE values across all classes and achieves the lowest mean RMSE of 0.1327, while MPGU-MLM also provides competitive performance. These results demonstrate the improved stability and accuracy of the proposed framework in abundance estimation. Regarding reconstruction accuracy, although some baseline methods obtain relatively low RE values, MPGU-MLM achieves the lowest RE, while MPGU-PPNM remains competitive. In particular, MPGU-MLM attains the smallest RE of 0.0119, confirming the effectiveness of the proposed method in preserving spectral fidelity under complex nonlinear mixing scenarios.

Figs. 9 and 10 provide visual comparisons of the estimated endmember spectra and abundance maps. As shown in Fig. 9, baseline methods exhibit varying degrees of spectral deviation from the ground truth, particularly for the weed and soil classes, where nonlinear effects are more pronounced. In contrast, MPGU-PPNM and MPGU-MLM generate spectral curves that closely match the reference across all materials, effectively preserving both overall trends and fine spectral details. As illustrated in Fig. 10, although most baseline methods can capture the general spatial distribution, their abundance maps often suffer from noise, fragmented regions, or blurred boundaries. In comparison, the proposed methods produce cleaner and more structured abundance maps, with clearer material boundaries and improved spatial consistency.

2) Results on the Jasper Ridge Dataset: Table III reports a quantitative comparison of different unmixing algorithms on the Jasper Ridge dataset. For SAD, several baseline methods exhibit class-dependent performance. For instance, MSNet achieves the best result on the water class, while some methods obtain relatively lower errors on specific materials. However, these advantages are limited to individual endmembers, and their overall SAD values remain higher. In contrast, MPGU-PPNM and MPGU-MLM consistently achieve lower SAD values across most classes, with MPGU-MLM attaining the lowest Mean SAD, demonstrating more accurate and stable endmember estimation. In terms of RMSE, MSNet performs best on tree and water, indicating that some baselines can achieve good performance under specific conditions. However, their performance varies significantly across different materials. In comparison, the proposed MPGU-based methods achieve more balanced results, with MPGU-PPNM obtaining

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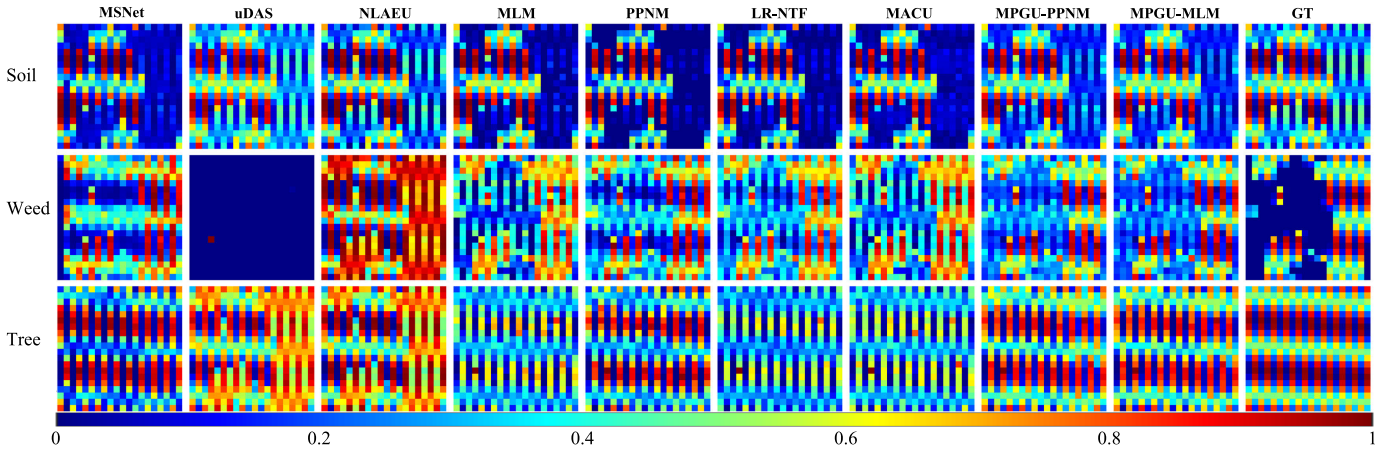


Fig. 10. Visual comparison of the abundance maps obtained by different hyperspectral unmixing methods on the Ray Tracing dataset. GT denotes the ground-truth abundances.

TABLE III

PERFORMANCE EVALUATION OF DIFFERENT UNMIXING METHODS ON THE JASPER RIDGE DATASET. BEST RESULTS ARE SHOWN IN BOLD, AND SECOND-BEST RESULTS ARE UNDERLINED. NOTE THAT MLM, PPNM, AND LR-NTF SHARE IDENTICAL VCA-BASED INITIALIZATION, LEADING TO IDENTICAL ENDMEMBER ESTIMATES

Methods		MSNet	uDAS	NLAEU	MLM	PPNM	LR-NTF	MACU	MPGU-PPNM	MPGU-MLM
SAD	Tree	0.1303	0.1677	0.1484	0.2649	0.2649	0.2649	0.1634	0.0696	**0.0430**
	Water	**0.0477**	0.1210	0.2633	0.2542	0.2542	0.2542	0.5010	0.0635	0.0611
	Soil	0.0623	0.1295	0.1171	0.2263	0.2263	0.2263	0.0855	0.0511	**0.0482**
	Road	0.0425	0.0546	0.0898	0.5360	0.5360	0.5360	0.2587	**0.0331**	0.0376
Mean SAD		0.0707	0.1182	0.1547	0.3203	0.3203	0.3203	0.2521	0.0543	**0.0475**
RMSE	Tree	**0.0676**	0.1532	0.1689	0.1746	0.2257	0.3583	0.3382	0.0939	0.1006
	Water	**0.0307**	0.1968	0.1901	0.0803	0.1686	0.4374	0.2368	0.0359	0.0313
	Soil	0.1318	0.1249	0.1772	0.2844	0.2069	0.2420	0.2121	**0.1084**	0.1135
	Road	0.1034	0.1217	0.1343	0.2487	0.3121	0.2851	0.1882	**0.0834**	0.0848
Mean RMSE		0.0834	0.1492	0.1676	0.1970	0.2283	0.3307	0.2438	**0.0804**	0.0826
RE		0.0676	0.0720	0.0807	0.2202	0.1447	0.1174	0.3451	**0.0674**	0.0679

the lowest Mean RMSE and showing strong performance on soil and road, while MPGU-MLM remains competitive across all classes. These results reflect the robustness of the proposed framework in abundance estimation. Regarding reconstruction accuracy, traditional nonlinear models (MLM, PPNM, and LR-NTF) yield relatively higher RE values. MSNet and the proposed methods obtain comparable RE values, with MPGU-PPNM achieving the lowest RE. This demonstrates the effectiveness of the proposed method in preserving spectral information under complex real scenarios.

In addition to the quantitative evaluation, we also provide visual comparisons of the estimated endmember spectra and abundance maps. As shown in Fig. 11, several baseline methods exhibit noticeable spectral deviations from the ground truth, particularly for soil and road, where spectral variations are more complex. In contrast, MPGU-PPNM and MPGU-MLM produce spectral signatures that closely match the reference across all materials, effectively preserving both global trends and fine spectral details. For the abundance maps in Fig. 12, the baseline methods generally reproduce the coarse distribution of materials but lose spatial consistency, especially in transitional regions such as the boundaries between soil and road. In comparison, MPGU-PPNM and MPGU-MLM not only capture the global spatial information more accurately but also preserve structural details, resulting in abundance maps that are much closer to the reference.

3) Results on the Washington DC Mall Dataset: Table IV reports the quantitative unmixing results on the Washington DC Mall dataset in terms of SAD, RMSE, and RE. In terms of endmember estimation accuracy, the proposed MPGU-PPNM and MPGU-MLM achieve the lowest Mean SAD values of 0.1742 and 0.1624, respectively, outperforming all baseline methods. In particular, both MPGU-based methods consistently yield lower SAD values for several materials such as Grass, Tree, and Trail where competing approaches often suffer from larger spectral angle errors. In terms of

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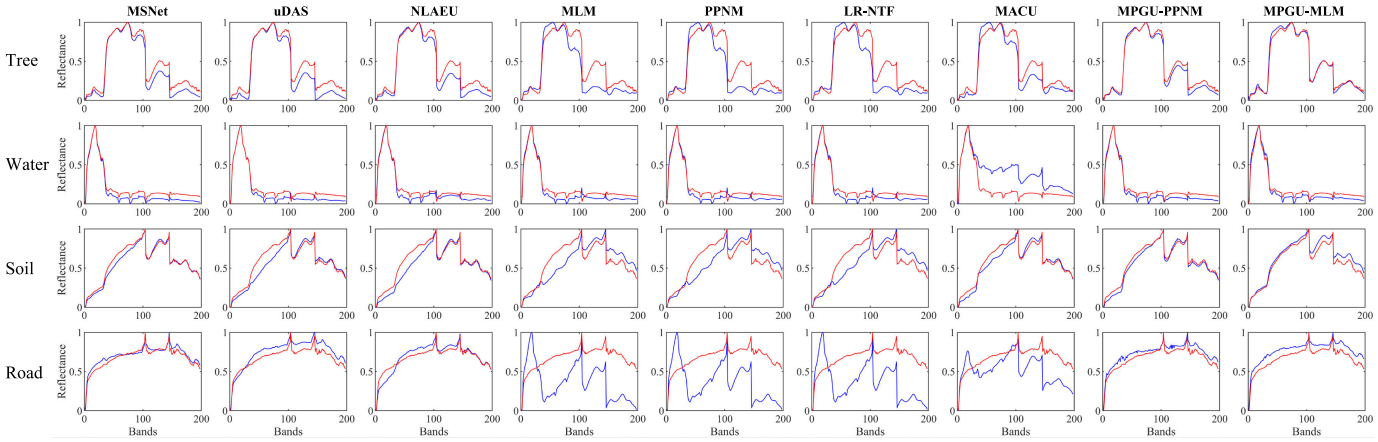


Fig. 11. Visual comparison of the endmembers obtained by different unmixing methods on the Jasper Ridge dataset. Blue curves denote the estimated endmembers, and the red curves denote the ground-truth endmembers. Note that MLM, PPNM, and LR-NTF share identical VCA-based initialization, leading to identical endmember estimates.

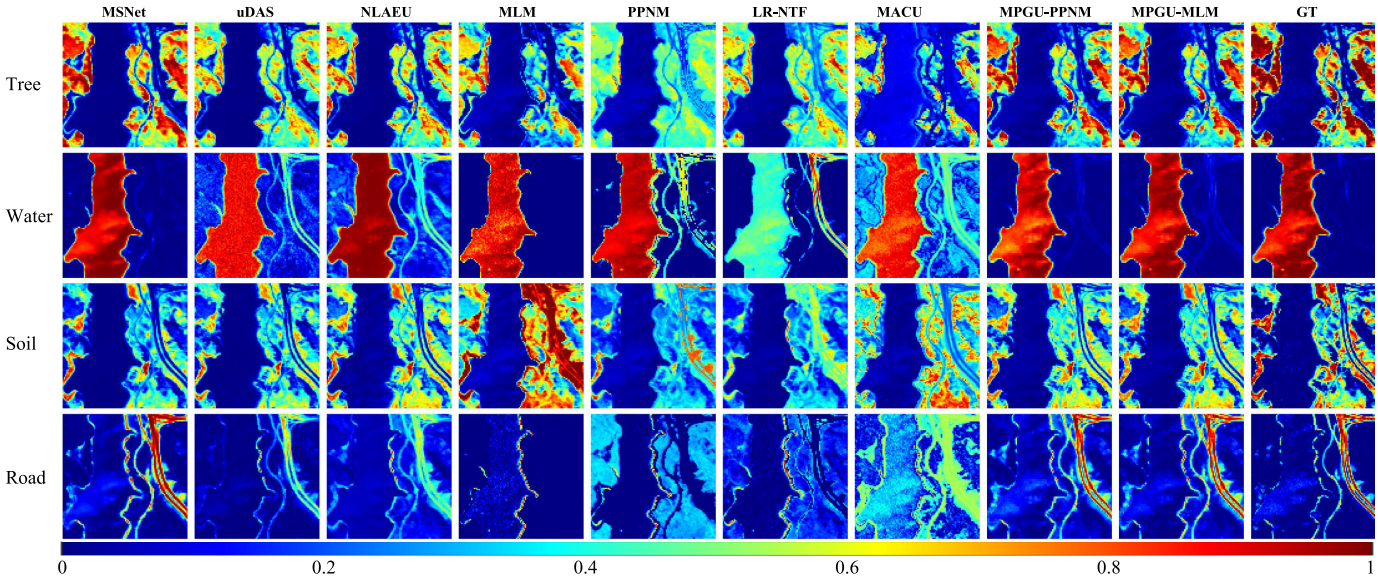


Fig. 12. Visual comparison of the abundance maps obtained by different hyperspectral unmixing methods on the Jasper Ridge dataset. GT denotes the ground-truth abundances.

abundance estimation, the proposed methods do not yield the best Mean RMSE. The lowest Mean RMSE is achieved by MSNet, followed by LR-NTF and PPNM, while MPGU-MLM and MPGU-PPNM obtain 0.2071 and 0.2133, respectively. Even so, MPGU-MLM achieves the lowest RMSE for Water, suggesting that the proposed framework remains effective for some difficult materials under complex mixing conditions. Regarding reconstruction performance, MPGU-MLM achieves the lowest RE value of 0.0338, while MPGU-PPNM also provides competitive results. Overall, these results demonstrate that the proposed MPGU framework achieves a favorable balance between endmember extraction, abundance estimation, and reconstruction performance in this more complex real-world scenario.

In addition to the quantitative evaluation, Figs. 13 and 14 illustrate the visual comparison of the estimated endmember spectra and abundance maps. From Fig. 13, it can be observed that several baseline methods show noticeable mismatches with the ground truth in certain spectral regions, especially for Tree and Road, where the spectral profiles are more complex. In contrast, MPGU-PPNM and MPGU-MLM provide more accurate spectral estimates, with closer alignment to the reference curves across most bands. As shown in Fig. 14, while baseline methods generally capture the main spatial patterns, their results tend to be less coherent in regions with complex material distributions. In comparison, the proposed methods yield more organized abundance representations, where the spatial structures of different materials are better preserved and more consistent with the ground truth. It is also worth noting that the advantage of the proposed methods becomes less pronounced for some challenging materials in the Washington DC dataset. For example, the abundance maps of Tree and Road in Fig. 14 still show local discrepancies compared with the ground truth. This may be attributed to the strong intraclass variability of vegetation and man-made materials, the spectral similarity among urban components,

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TABLE IV

PERFORMANCE EVALUATION OF DIFFERENT UNMIXING METHODS ON THE WASHINGTON DC MALL DATASET. BEST RESULTS ARE SHOWN IN BOLD, AND SECOND-BEST RESULTS ARE UNDERLINED. NOTE THAT MLM, PPNM, AND LR-NTF SHARE IDENTICAL VCA-BASED INITIALIZATION, LEADING TO IDENTICAL ENDMEMBER ESTIMATES

Washington DC Mall		MSNet	uDAS	NLAEU	MLM	PPNM	LR-NTF	MACU	MPGU-PPNM	MPGU-MLM
SAD	Grass	0.1064	0.2045	0.1907	0.2009	0.2009	0.2009	0.2138	**0.0962**	0.1257
	Tree	0.3110	0.1925	0.3506	0.3323	0.3323	0.3323	0.3952	**0.0973**	0.1892
	Road	0.3162	**0.2126**	0.2471	0.2400	0.2400	0.2400	0.2553	0.5922	0.4048
	Water	**0.0322**	0.2549	0.1537	0.0816	0.0816	0.0816	0.0652	0.0403	0.0546
	Trail	0.1931	0.3940	0.1801	0.1705	0.1705	0.1705	0.1703	0.0453	**0.0378**
Mean SAD		0.1918	0.2517	0.2244	0.2050	0.2050	0.2050	0.2199	0.1742	**0.1624**
RMSE	Grass	**0.1140**	0.3079	0.3532	0.1876	0.1856	0.1766	0.2531	0.1510	0.1246
	Tree	0.2378	0.3629	**0.2553**	0.2569	0.2381	0.2567	0.2777	0.3638	0.4417
	Road	0.2076	0.2326	0.2049	0.2231	**0.2008**	0.2080	0.2170	0.2909	0.2127
	Water	0.2122	0.2547	0.3216	0.2706	0.2781	0.2551	0.3458	0.1566	**0.1422**
	Trail	**0.0846**	0.1790	0.1292	0.1287	0.1063	0.0921	0.0881	0.1043	0.1144
Mean RMSE		**0.1712**	0.2674	0.2528	0.2134	0.2018	0.1977	0.2364	0.2133	0.2071
RE		0.0379	0.1266	0.0498	0.0397	0.0385	0.0352	0.0517	0.0350	**0.0338**

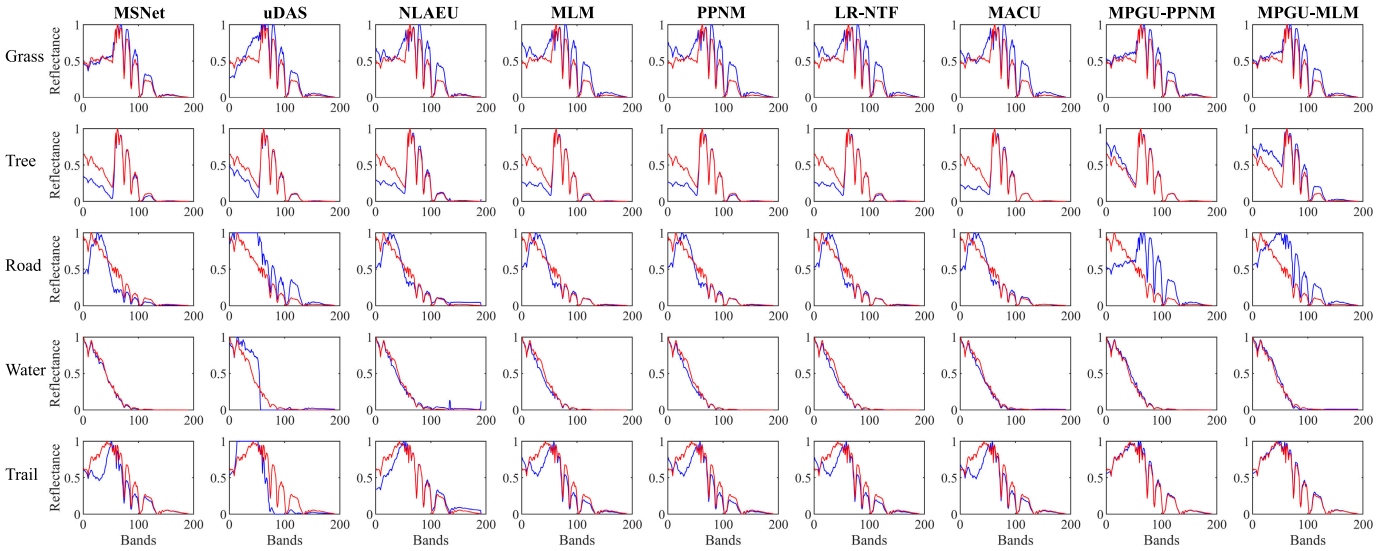


Fig. 13. Visual comparison of the endmembers obtained by different unmixing methods on the Washington DC Mall dataset. Blue curves denote the estimated endmembers, and the red curves denote the ground-truth endmembers. Note that MLM, PPNM, and LR-NTF share identical VCA-based initialization, leading to identical endmember estimates.

shadow interference, and complex nonlinear mixing effects. Therefore, although MPGU-PPNM and MPGU-MLM provide more spatially organized abundance representations, further improvement is still needed to enhance their robustness in highly heterogeneous real urban scenarios.

F. Efficiency Evaluation

Table V reports the computational cost of different unmixing methods on both synthetic and real datasets. Traditional nonlinear models, such as PPNM, MLM, and MACU, generally incur high computational costs. In particular, LR-NTF exhibits consistently large running times across all datasets due to its iterative tensor factorization scheme, making it less suitable for practical applications. Deep-learning-based methods such as uDAS and MSNet offer improved efficiency compared to traditional approaches; however, their computational cost still increases in more complex scenarios, especially on real datasets. Although NLAEU achieves relatively competitive running times, its unmixing performance remains limited. In contrast, the proposed MPGU-PPNM and MPGU-MLM demonstrate consistently efficient inference across both synthetic and real datasets. On synthetic datasets, both methods achieve low running times while maintaining strong performance. On real datasets, although the computational cost increases due to higher data complexity, the proposed methods

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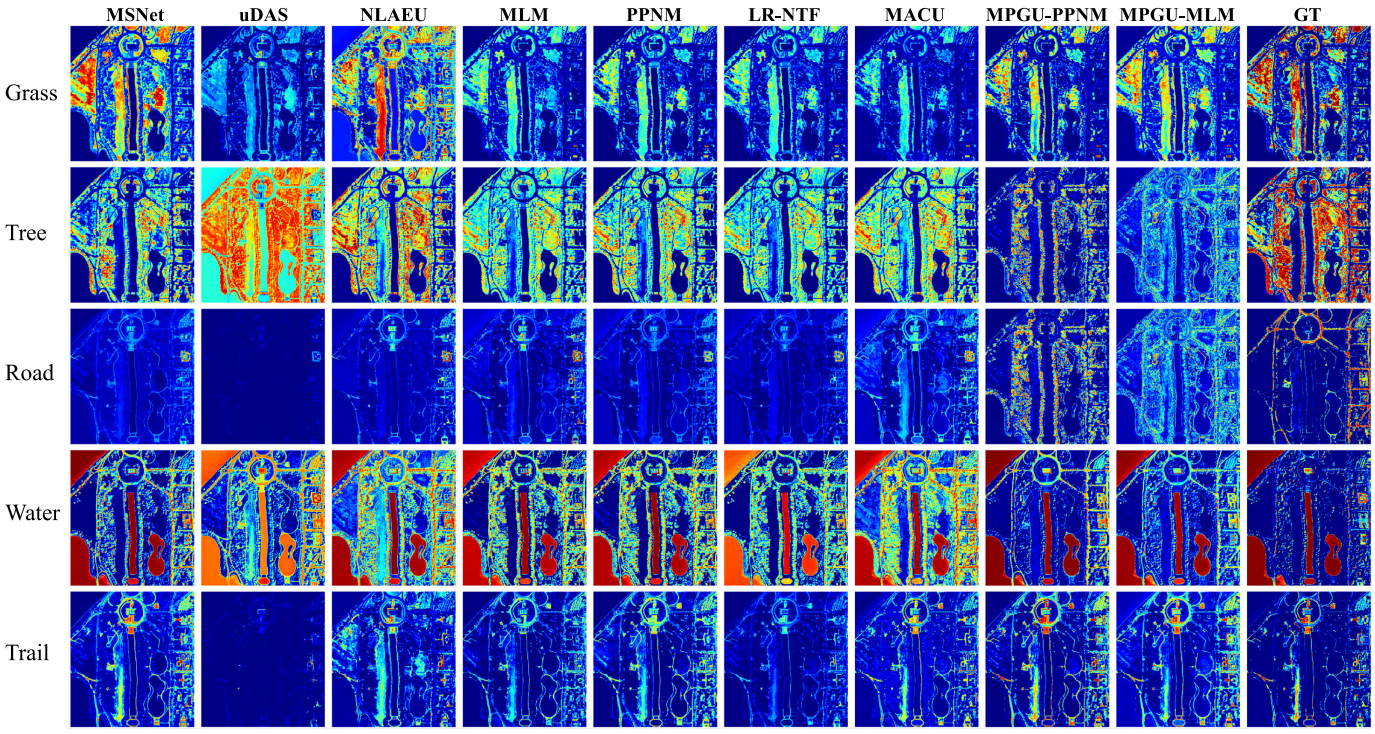


Fig. 14. Visual comparison of the abundance maps obtained by different hyperspectral unmixing methods on the Washington DC Mall dataset. GT denotes the ground-truth abundances.

TABLE V
COMPARISON OF THE COMPUTATIONAL COST OF THE METHODS ON DIFFERENT DATASETS (RUNNING TIME IN SECONDS)

Methods		MSNet	uDAS	NLAEU	MLM	PPNM	LR-NTF	MACU	MPGU-PPNM	MPGU-MLM
Synthetic datasets	PPNM-based	39.67	9.26	5.60	34.42	57.26	412.87	89.77	8.58	6.73
	MLM-based	38.66	10.53	5.56	47.51	48.01	197.52	84.22	9.00	11.44
Real datasets	Ray Tracing	43.56	8.37	2.33	0.75	0.41	1.60	15.97	3.71	3.70
	Jasper Ridge	52.53	8.36	5.06	9.37	16.78	5.14	75.75	4.12	4.15
	Washington DC Mall	25.4837	508.3033	45.3925	140.7064	106.3336	159.4321	523.2159	21.1697	17.9369

still remain significantly more efficient than most baseline approaches. Overall, these results indicate that the proposed MPGU framework achieves a favorable balance between computational efficiency and unmixing performance, making it well-suited for practical hyperspectral applications.

G. Hyperparameter Analysis

To investigate the robustness of the proposed framework with respect to the regularization parameters in the objective function, we conduct a hyperparameter analysis for both MPGU-PPNM and MPGU-MLM on representative synthetic datasets. The influence of the geometric regularization parameter β, the angular consistency parameter γ, and the latent regularization parameter λ is evaluated in terms of Mean SAD, Mean RMSE, and RE. The corresponding results are reported in Fig. 15. We first analyze the sensitivity of MPGU-PPNM on the PPNM-based synthetic dataset with 30-dB noise, as shown in Fig. 15(a)–(c). For the PPNM-based synthetic dataset, the best performance is obtained at $\beta = 10^{-6}$, $\gamma = 10^{-5}$, and $\lambda = 10^{-2}$. Small values of β provide an appropriate balance between geometric regularization and reconstruction accuracy, while overly large β leads to a clear performance degradation. Moderate values of γ and λ further improve both endmember and abundance estimation accuracy. We further analyze the hyperparameter sensitivity of MPGU-MLM on the MLM-based synthetic dataset, as illustrated in Fig. 15(d)–(f). For the MLM-based synthetic dataset with 30-dB noise, the lowest estimation errors are obtained at $\beta = 10^{-6}$, $\gamma = 10^{-5}$, and $\lambda = 0.1$. The performance is relatively stable for small values of β, while large β values lead to obvious performance degradation. Moderate γ provides the best tradeoff between spectral coherence and flexibility, and $\lambda = 0.1$ yields the most stable reconstruction performance.

V. DISCUSSION

A. Evaluation of the Effectiveness of the LST

To explicitly analyze the role of the proposed LST strategy and its incorporation into the MSVC loss function, we conduct an ablation study considering three distinct situations: not using MSVC at all, applying MSVC on the network-based LR

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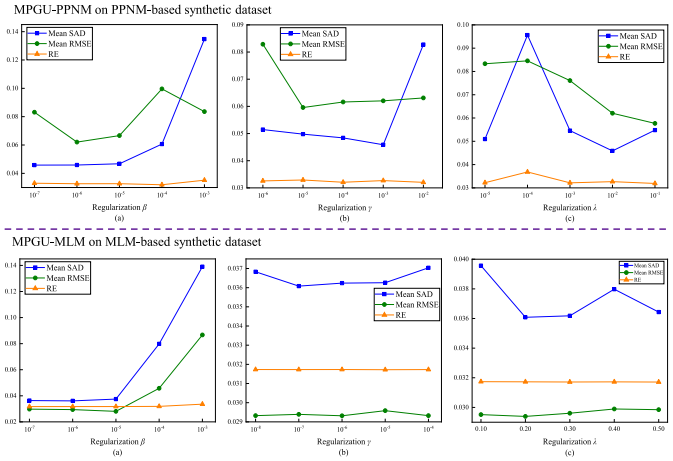


Fig. 15. Hyperparameter analysis conducted on the proposed MPGU-PPNM and MPGU-MLM by varying: regularization parameter β, regularization parameter γ, and regularization parameter λ for different datasets: (a)–(c) PPNM-based synthetic dataset and (d)–(f) MLM-based synthetic dataset.

TABLE VI

PERFORMANCE OF THE PROPOSED NETWORK, WITHOUT APPLYING MSVC AND WITH APPLYING MSVC ON $\hat{\mathbf{Y}}^{\text{lin}}$ AND X, RESPECTIVELY, ON PPNM-BASED AND MLM-BASED SYNTHETIC DATASETS AT DIFFERENT NOISE LEVELS. BEST RESULTS ARE SHOWN IN BOLD

MSVC	$\hat{\mathbf{Y}}^{\text{lin}}$	X	MPGU-PPNM (PPNM-based data)				MPGU-MLM (MLM-based data)			
			40 dB		50 dB		40 dB		50 dB	
			SAD	RMSE	SAD	RMSE	SAD	RMSE	SAD	RMSE
Without	–	–	0.1047	0.0679	0.0886	0.0727	0.0402	0.0294	0.0309	0.0247
With	✓	×	0.0532	0.0501	0.0530	0.0520	0.0392	0.0299	0.0336	0.0278
	×	✓	**0.0368**	**0.0282**	**0.0336**	**0.0313**	**0.0372**	**0.0284**	**0.0308**	**0.0244**

$\hat{\mathbf{Y}}^{\text{lin}}$ and applying MSVC on the LST-derived linear component $\mathbf{X}$. The results are summarized in Table VI.

When MSVC is not applied ("Without"), the model relies solely on NR, resulting in relatively high SAD and RMSE values, especially under lower SNR conditions. When MSVC is enabled ("With"), two different linear supervision strategies are compared. The first one uses $\hat{\mathbf{Y}}^{\text{lin}}$, which is obtained by combining the estimated endmembers $\hat{\mathbf{M}}$ and abundances $\hat{\mathbf{A}}$. Since $\hat{\mathbf{Y}}^{\text{lin}}$ depends on both estimates, its accuracy is inevitably affected by errors in endmember and abundance estimation. The second strategy uses the proposed LST-derived linear component $\mathbf{X}$, which is computed through the proposed LST using the observed hyperspectral image $\mathbf{Y}$ and the estimated nonlinear parameters $\hat{\mathbf{P}}$, and is therefore independent of endmember and abundance estimation errors.

From Table VI, it can be observed that introducing MSVC already brings noticeable performance improvements compared with one that does not use MSVC. More importantly, using $\mathbf{X}$ as the linear reference for the MSVC loss consistently yields lower SAD and RMSE values compared with using $\hat{\mathbf{Y}}^{\text{lin}}$ across different noise levels and for both PPNM-based and MLM-based data. Although acceptable performance can occasionally be obtained without linear supervision at very high SNR levels, incorporating the LST-derived linear component $\mathbf{X}$ consistently provides more accurate and robust results across different noise conditions. These results demonstrate that the LST-derived linear component $\mathbf{X}$ serves as a more reliable and less biased reference for enforcing the MSVC constraint, thereby validating the effectiveness of the proposed LST strategy and its integration into the loss function.

B. Performance on Linear Mixing Data

Although the proposed MPGU-Net is designed for nonlinear hyperspectral unmixing, it is also important to examine its behavior under purely linear mixing conditions. To this end, we conduct additional experiments on synthetic datasets generated according to the LMM while keeping the same experimental configuration as the nonlinear datasets (i.e., identical endmembers, abundance distributions, and noise settings), such that the only difference lies in the mixing mechanism. Specifically, the proposed framework is evaluated under two reconstruction strategies during training: 1) NR, which corresponds to the default setting of MPGU-Net and 2) LR, where the reconstructed spectra are obtained through the linear combination of the estimated endmembers and abundances. This comparison aims to investigate how the reconstruction mechanism influences performance when the data strictly follows a linear model.

Table VII presents the comparison between linear (LR) and nonlinear (NR) reconstruction strategies under the LMM. It can be observed that the LR strategy generally yields more stable endmember estimation in terms of SAD, particularly at higher SNR levels, which is consistent with the linear nature of the data, whereas the NR strategy often achieves comparable or even lower REs due to its greater flexibility in fitting the observed spectra. Nevertheless, the NR strategy remains highly competitive across all metrics and SNR conditions, while showing different advantages depending on the evaluation metric. This indicates that the nonlinear modeling in MPGU-Net does not compromise its effectiveness when applied to linear mixtures, and is capable of approximating linear interactions with sufficient accuracy. Overall, these results demonstrate that the proposed nonlinear framework maintains strong robustness and generalization ability, even when applied to purely linear data, without requiring explicit adaptation to the LMM.

C. Effect of Abundance Sparsity

To evaluate the performance of the proposed method under different abundance sparsity conditions, we conduct an additional experiment by varying the number of active endmembers in each pixel. Specifically, the number of active endmembers is controlled from 1 to 6, corresponding to different sparsity levels. For each setting, the abundances are generated such that only a fixed number of endmembers is active in each pixel, while the remaining ones are set to zero. The active endmembers are uniformly sampled from the set of six endmembers to avoid bias toward specific materials. The experiments are conducted on the PPNM-based and MLM-based synthetic datasets using MPGU-PPNM and MPGU-MLM, respectively.

The quantitative results are reported in Table VIII. When only one endmember is active, the presence of pure pixels simplifies the unmixing task, resulting in the best performance.

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TABLE VII
COMPARISON OF LINEAR (LR) AND NONLINEAR (NR) RECONSTRUCTION STRATEGIES ON LMM-BASED SYNTHETIC DATASET (BEST RESULTS ARE SHOWN IN BOLD)

SNR (dB)		20			30			40			50		
Metrics		SAD	RMSE	RE	SAD	RMSE	RE	SAD	RMSE	RE	SAD	RMSE	RE
MPGU-PPNM	LR	**0.0595**	**0.0624**	0.0994	**0.0428**	**0.0493**	0.0385	**0.0301**	0.0351	**0.0206**	**0.0255**	**0.0247**	0.0117
	NR	0.0728	0.0642	**0.0991**	0.0457	0.0546	**0.0316**	0.0453	**0.0289**	**0.0103**	0.0334	0.0507	0.0048
MPGU-MLM	LR	**0.0582**	**0.0627**	0.0999	**0.0398**	0.0525	0.0336	**0.0320**	0.0518	**0.0145**	**0.0137**	**0.0204**	0.0043
	NR	0.0832	0.0774	**0.0992**	0.0536	**0.0435**	**0.0316**	0.0415	**0.0352**	0.0106	0.0370	0.0349	**0.0050**

TABLE VIII
UNMIXING PERFORMANCE UNDER DIFFERENT NUMBERS OF ACTIVE ENDMEMBERS

Active endmembers	MPGU-PPNM			MPGU-MLM		
	SAD	RMSE	RE	SAD	RMSE	RE
1	0.0072	0.0291	0.0353	0.0236	0.0298	0.0476
2	0.0363	0.0769	0.0337	0.0483	0.0875	0.0453
3	0.0567	0.0726	0.0328	0.0576	0.0895	0.0445
4	0.0514	0.0921	0.0327	0.0584	0.0581	0.0437
5	0.0529	0.0769	0.0328	0.0604	0.0541	0.0442
6 (Ours)	0.0459	0.0620	0.0327	0.0361	0.0294	0.0317

As the number of active endmembers increases, the unmixing task becomes more challenging due to more complex spectral interactions, leading to a gradual performance degradation. Nevertheless, even when all endmembers are active, the proposed method maintains stable and reliable performance. These results demonstrate the robustness of the proposed framework under varying sparsity conditions.

D. Spectral Variability

Endmember variability is ubiquitous in real hyperspectral scenes and remains a challenging issue. When the measured spectra are scaled versions of the true reflectance spectra, e.g., due to variations in illumination conditions, topography, etc., the linear components obtained via LST are approximately scaled versions of the true linear components, by rescaling the nonlinear parameters [see (7) and (9)]. Since the method minimizes the spectral angle loss between the estimated linear components and those determined by LST, it is less sensitive to such scaling effects in the measured spectra. However, when endmember variability is more complex, the estimated linear components become less reliable, which may lead to inaccurate endmember estimation.

VI. CONCLUSION

This article presents MPGU-Net, a physics-guided end-to-end deep learning framework designed for nonlinear hyperspectral unmixing in no-pure-pixel scenarios. The proposed MPGU-Net integrates two representative nonlinear mixing models (i.e., PPNM and MLM) directly into the reconstruction module. This allows the network to capture a broad spectrum of nonlinear interactions with improved accuracy and physical interpretability. In addition, a nonlinear-to-linear space transformation strategy is introduced to approximate the underlying linear structure of nonlinear mixtures, enabling the application of geometric constraints for endmember estimation that are not valid under standard nonlinear conditions. The overall framework includes a variational endmember generator network, an abundance estimation network, a nonlinearity parameter estimation network, and a spectral reconstruction network that jointly optimize endmembers, abundances, and reconstructed spectra. Extensive experiments on both synthetic and real hyperspectral datasets confirm that MPGU-Net achieves superior performance compared to existing linear and nonlinear unmixing methods. The proposed approach yields more accurate endmember signatures, more spatially coherent abundance maps, and higher reconstruction accuracy. Furthermore, MPGU-Net does not rely on the presence of pure pixels, which makes it suitable for real-world applications where ideal conditions are often not met.

Future work may explore more adaptive alternatives to the current LST, particularly for scenarios involving unknown or highly complex nonlinear mixing behaviors. While the adopted LST effectively restores geometric constraints under specific models, its analytical form may limit flexibility in general-purpose unmixing tasks. Learning-based or model-agnostic approximations of the linear space could offer a promising direction for further improving endmember regularization under diverse nonlinear conditions. In addition, further efforts will be devoted to improving the robustness of MPGU-Net in highly heterogeneous real-world scenarios, where strong spectral variability, shadow effects, and complex spatial distributions may still affect the abundance estimation of challenging materials.

REFERENCES

[1] M. J. Mashala, T. Dube, B. T. Mudereri, K. K. Ayisi, and M. R. Ramudzuli, "A systematic review on advancements in remote sensing for assessing and monitoring land use and land cover changes impacts on surface water resources in semi-arid tropical environments," *Remote Sens.*, vol. 15, no. 16, p. 3926, Aug. 2023.

[2] H. Shirmard, E. Farahbakhsh, R. D. Müller, and R. Chandra, "A review of machine learning in processing remote sensing data for mineral exploration," *Remote Sens. Environ.*, vol. 268, Jan. 2022, Art. no. 112750.

[3] K. E. Joyce, S. E. Belliss, S. V. Samsonov, S. J. McNeill, and P. J. Glassey, "A review of the status of satellite remote sensing and image processing techniques for mapping natural hazards and disasters," *Prog. Phys. Geography, Earth Environ.*, vol. 33, no. 2, pp. 183–207, Apr. 2009.

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[4] J. Yang et al., "The role of satellite remote sensing in climate change studies," *Nature Climate Change*, vol. 3, no. 10, pp. 875–883, Oct. 2013.

[5] Y. Xu et al., "A comb concatenation diffusion model for hyperspectral image super-resolution," *Eng. Appl. Artif. Intell.*, vol. 160, Nov. 2025, Art. no. 111985.

[6] Y. Fu, J. Lyu, P. Ma, Z. Liu, and M. K. Ng, "A unified SAM-guided self-prompt learning framework for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5008014.

[7] Z. Han, J. Yang, L. Gao, Z. Zeng, B. Zhang, and J. Chanussot, "Subpixel spectral variability network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504014.

[8] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[9] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[10] R. Heylen, M. Parente, and P. Gader, "A review of nonlinear hyperspectral unmixing methods," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 1844–1868, Jun. 2014.

[11] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[12] B. Rasti, A. Zouaoui, J. Mairal, and J. Chanussot, "Image processing and machine learning for hyperspectral unmixing: An overview and the HySUPP Python package," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517631.

[13] Z. Han, D. Hong, L. Gao, J. Yao, B. Zhang, and J. Chanussot, "Multimodal hyperspectral unmixing: Insights from attention networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524913.

[14] J. Wei and X. Wang, "An overview on linear unmixing of hyperspectral data," *Math. Problems Eng.*, vol. 2020, no. 1, Aug. 2020, Art. no. 3735403.

[15] P.-A. Thouvenin, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral unmixing with spectral variability using a perturbed linear mixing model," *IEEE Trans. Signal Process.*, vol. 64, no. 2, pp. 525–538, Jan. 2016.

[16] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[17] J. M. P. Nascimento and J. M. Bioucas-Dias, "Nonlinear mixture model for hyperspectral unmixing," *Proc. SPIE*, vol. 7477, pp. 157–164, Apr. 2009.

[18] J. M. P. Nascimento and G. Martín, "Nonlinear spectral unmixing," in *Data Handling in Science and Technology*, vol. 32. Amsterdam, The Netherlands: Elsevier, 2019, pp. 151–166.

[19] J. M. P. Nascimento and J. M. B. Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[20] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," *Proc. SPIE*, vol. 3753, pp. 266–275, May 1999.

[21] X. Tao, T. Cui, and P. Ren, "Cofactor-based efficient endmember extraction for green algae area estimation," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 6, pp. 849–853, Jun. 2019.

[22] D. Schillemans, R. Tauler, M. Haverkorn, G. H. Tinnevelt, J. J. Jansen, and M. Ghaffari, "Enhancing chemical image analysis: Utilizing the successive projection algorithm for unmixing," *Spectrochimica Acta A, Mol. Biomolecular Spectrosc.*, vol. 338, Oct. 2025, Art. no. 126201.

[23] L. Miao and H. Qi, "Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 3, pp. 765–777, Mar. 2007.

[24] J. Li, A. Agathos, D. Zaharie, J. M. Bioucas-Dias, A. Plaza, and X. Li, "Minimum volume simplex analysis: A fast algorithm for linear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 5067–5082, Sep. 2015.

[25] J. M. Bioucas-Dias, "A variable splitting augmented Lagrangian approach to linear spectral unmixing," in *Proc. 1st Workshop Hyperspectral Image Signal Process., Evol. Remote Sens.*, Aug. 2009, pp. 1–4.

[26] N. Dobigeon, S. Moussaoui, M. Coulon, J.-Y. Tourneret, and A. O. Hero, "Joint Bayesian endmember extraction and linear unmixing for hyperspectral imagery," *IEEE Trans. Signal Process.*, vol. 57, no. 11, pp. 4355–4368, Nov. 2009.

[27] K. E. Themelis, A. A. Rontogiannis, and K. D. Koutroumbas, "A novel hierarchical Bayesian approach for sparse semisupervised hyperspectral unmixing," *IEEE Trans. Signal Process.*, vol. 60, no. 2, pp. 585–599, Feb. 2012.

[28] H. Liu, Y. Lu, Z. Wu, Q. Du, J. Chanussot, and Z. Wei, "Bayesian unmixing of hyperspectral image sequence with composite priors for abundance and endmember variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[29] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Collaborative sparse regression for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 341–354, Jan. 2014.

[30] S. Zhang, J. Li, K. Liu, C. Deng, L. Liu, and A. Plaza, "Hyperspectral unmixing based on local collaborative sparse regression," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 631–635, May 2016.

[31] C. Li, Y. Ma, X. Mei, C. Liu, and J. Ma, "Hyperspectral unmixing with robust collaborative sparse regression," *Remote Sens.*, vol. 8, no. 7, p. 588, Jul. 2016.

[32] B. Rasti and B. Koirala, "SUnCNN: Sparse unmixing using unsupervised convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[33] L. Gao, Z. Wang, L. Zhuang, H. Yu, B. Zhang, and J. Chanussot, "Using low-rank representation of abundance maps and nonnegative tensor factorization for hyperspectral nonlinear unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2021.

[34] Y. Hashemi-Nazari, A. Tajaddini, F. Saberi-Movahed, F. Alonso-Fernandez, and P. Tiwari, "Robust oblique projection and weighted NMF for hyperspectral unmixing," *Pattern Recognit.*, vol. 170, Feb. 2026, Art. no. 112029.

[35] S. Shu, T.-Z. Huang, J. Huang, and G. Vivone, "Nesterov-accelerated non-negative matrix factorization unrolling network for hyperspectral unmixing," *Neurocomputing*, vol. 653, Nov. 2025, Art. no. 131202.

[36] B. Yang, W. Luo, and B. Wang, "Constrained nonnegative matrix factorization based on particle swarm optimization for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3693–3710, Aug. 2017.

[37] Y. Miao and B. Yang, "Multilevel reweighted sparse hyperspectral unmixing using superpixel segmentation and particle swarm optimization," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[38] Y. Miao and B. Yang, "Sparse unmixing for hyperspectral imagery via comprehensive-learning-based particle swarm optimization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9727–9742, 2021.

[39] Y. Altmann, N. Dobigeon, and J.-Y. Tourneret, "Bilinear models for nonlinear unmixing of hyperspectral images," in *Proc. 3rd Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Lisbon, Portugal, Jun. 2011, pp. 1–4.

[40] W. Luo, L. Gao, R. Zhang, A. Marinoni, and B. Zhang, "Bilinear normal mixing model for spectral unmixing," *IET Image Process.*, vol. 13, no. 2, pp. 344–354, Feb. 2019.

[41] A. Halimi, Y. Altmann, N. Dobigeon, and J.-Y. Tourneret, "Nonlinear unmixing of hyperspectral images using a generalized bilinear model," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4153–4162, Nov. 2011.

[42] C. Li et al., "Hyperspectral unmixing with bandwise generalized bilinear model," *Remote Sens.*, vol. 10, no. 10, p. 1600, Oct. 2018.

[43] Y. Altmann, A. Halimi, N. Dobigeon, and J.-Y. Tourneret, "Supervised nonlinear spectral unmixing using a postnonlinear mixing model for hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 21, no. 6, pp. 3017–3025, Jun. 2012.

[44] Y. Altmann, N. Dobigeon, and J.-Y. Tourneret, "Nonlinearity detection in hyperspectral images using a polynomial post-nonlinear mixing model," *IEEE Trans. Image Process.*, vol. 22, no. 4, pp. 1267–1276, Apr. 2013.

[45] Y. Altmann, N. Dobigeon, and J.-Y. Tourneret, "Unsupervised post-nonlinear unmixing of hyperspectral images using a Hamiltonian Monte Carlo algorithm," *IEEE Trans. Image Process.*, vol. 23, no. 6, pp. 2663–2675, Jun. 2014.

[46] R. Heylen and P. Scheunders, "A multilinear mixing model for nonlinear spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 240–251, Jan. 2016.

[47] B. Yang and B. Wang, "Band-wise nonlinear unmixing for hyperspectral imagery using an extended multilinear mixing model," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6747–6762, Nov. 2018.

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[48] M. Li, B. Yang, and B. Wang, "Spectral–spatial reweighted robust nonlinear unmixing for hyperspectral images based on an extended multilinear mixing model," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5543917.

[49] D. U. Campos-Delgado et al., "Robust and unified semi-supervised unmixing of hyperspectral imaging for linear and multilinear models," *IEEE Access*, vol. 13, pp. 53140–53158, 2025.

[50] B. Hapke, "Bidirectional reflectance spectroscopy: 1. Theory," *J. Geophys. Res., Solid Earth*, vol. 86, no. B4, pp. 3039–3054, Apr. 1981.

[51] B. Rasti, B. Koirala, and P. Scheunders, "HapkeCNN: Blind nonlinear unmixing for intimate mixtures using Hapke model and convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536315.

[52] R. Ammanouil, A. Ferrari, C. Richard, and S. Mathieu, "Nonlinear unmixing of hyperspectral data with vector-valued kernel functions," *IEEE Trans. Image Process.*, vol. 26, no. 1, pp. 340–354, Jan. 2017.

[53] R. A. Borsoi, T. Imbiriba, J. C. M. Bermudez, and C. Richard, "A blind multiscale spatial regularization framework for kernel-based spectral unmixing," *IEEE Trans. Image Process.*, vol. 29, pp. 4965–4979, 2020.

[54] Y. Gu, S. Wang, and X. Jia, "Spectral unmixing in multiple-kernel Hilbert space for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 7, pp. 3968–3981, Jul. 2013.

[55] Y. Altmann, N. Dobigeon, J-Y. Tourneret, and S. McLaughlin, "Nonlinear unmixing of hyperspectral images using radial basis functions and orthogonal least squares," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2011, pp. 1151–1154.

[56] K. J. Guilfoyle, M. L. Althouse, and C.-I. Chang, "A quantitative and comparative analysis of linear and nonlinear spectral mixture models using radial basis function neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 10, pp. 2314–2318, Oct. 2001.

[57] A. M. Saranathan and M. Parente, "Unmixing multiple intimate mixtures via a locally low-rank representation," in *Proc. 8th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Aug. 2016, pp. 1–4.

[58] N. H. Nguyen, C. Richard, P. Honeine, and C. Theys, "Hyperspectral image unmixing using manifold learning methods derivations and comparative tests," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2012, pp. 3086–3089.

[59] X. Lu, H. Wu, Y. Yuan, P. Yan, and X. Li, "Manifold regularized sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2815–2826, May 2013.

[60] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[61] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[62] X. Tao et al., "A new deep convolutional network for effective hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6999–7012, 2022.

[63] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[64] X. Tao, B. Koirala, B. Rasti, A. Plaza, and P. Scheunders, "A spectral–spatial attention network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5516315.

[65] D. Bank, N. Koenigstein, and R. Giryes, "Autoencoders," in *Machine Learning for Data Science Handbook: Data Mining and Knowledge Discovery Handbook*. Cham, Switzerland: Springer, 2023, pp. 353–374.

[66] Z. Li, F. Liu, W. Yang, S. Peng, and J. Zhou, "A survey of convolutional neural networks: Analysis, applications, and prospects," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 6999–7019, Dec. 2022.

[67] A. Ajit, K. Acharya, and A. Samanta, "A review of convolutional neural networks," in *Proc. Int. Conf. Emerg. Trends Inf. Technol. Eng. (ic-ETITE)*, Feb. 2020, pp. 1–5.

[68] P. Ghosh, S. K. Roy, B. Koirala, B. Rasti, and P. Scheunders, "Hyperspectral unmixing using transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535116.

[69] L. R. Medsker et al., "Recurrent neural networks," *Des. Appl.*, vol. 5, nos. 64–67, p. 2, 2001.

[70] X. Tao, M. E. Paoletti, Z. Wu, J. M. Haut, P. Ren, and A. Plaza, "An abundance-guided attention network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505414.

[71] Y. Yu, Y. Ma, X. Mei, F. Fan, J. Huang, and H. Li, "Multi-stage convolutional autoencoder network for hyperspectral unmixing," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 113, Sep. 2022, Art. no. 102981.

[72] X. Tao, B. Koirala, A. Plaza, and P. Scheunders, "A new dual-feature fusion network for enhanced hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5540113.

[73] B. Rasti, B. Koirala, P. Scheunders, and J. Chanussot, "MiSiCNet: Minimum simplex convolutional network for deep hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522815.

[74] A. Ahmed, O. Duran, Y. Zweiri, and M. Smith, "Hybrid spectral unmixing: Using artificial neural networks for linear/non-linear switching," *Remote Sens.*, vol. 9, no. 8, p. 775, Jul. 2017.

[75] M. Zhao, M. Wang, J. Chen, and S. Rahardja, "Hyperspectral unmixing for additive nonlinear models with a 3-D-CNN autoencoder network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[76] M. Wang, M. Zhao, J. Chen, and S. Rahardja, "Nonlinear unmixing of hyperspectral data via deep autoencoder networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 9, pp. 1467–1471, Sep. 2019.

[77] X. Chen, X. Zhang, M. Ren, B. Zhou, Z. Feng, and J. Cheng, "An improved hyperspectral unmixing approach based on a spatial–spectral adaptive nonlinear unmixing network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 9680–9696, 2023.

[78] M. Zhao, L. Yan, and J. Chen, "LSTM-DNN based autoencoder network for nonlinear hyperspectral image unmixing," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 2, pp. 295–309, Feb. 2021.

[79] S. Shi, M. Zhao, L. Zhang, Y. Altmann, and J. Chen, "Probabilistic generative model for hyperspectral unmixing accounting for endmember variability," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[80] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.

[81] Y. Qu and H. Qi, "UDAS: An untied denoising autoencoder with sparsity for spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1698–1712, Mar. 2019.

[82] H. Li, R. A. Borsoi, T. Imbiriba, P. Closas, J. C. M. Bermudez, and D. Erdoğmuş, "Model-based deep autoencoder networks for nonlinear hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[83] B. Somers, L. Tits, and P. Coppin, "Quantifying nonlinear spectral mixing in vegetated areas: Computer simulation model validation and first results," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 1956–1965, Jun. 2014.

[84] J. Stuckens, B. Somers, S. Delalieux, W. W. Verstraeten, and P. Coppin, "The impact of common assumptions on canopy radiative transfer simulations: A case study in citrus orchards," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 110, nos. 1–2, pp. 1–21, Jan. 2009.

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